Multiple Hypothesis Testing in Group Sequential and Adaptive Clinical Trials

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Outline of course

1. **Group sequential tests**

   Error spending tests

   Example: A group sequential design with survival data

   Inference on termination

2. **Multiple testing procedures**

   Graphical representation of multiple testing procedures

   Combining multiple testing and group sequential tests

   Testing a secondary endpoint after a group sequential test

3. **Adaptive clinical trial designs**

   Combination tests and Closed Testing Procedures

   A Phase III trial comparing two treatments with a control

Part 1. Group sequential tests

1.1. Group sequential tests: Introduction

Suppose a new treatment (Treatment A) is to be compared to a placebo or positive control (Treatment B) in a Phase III trial.

The treatment effect θ for the **primary endpoint** represents the advantage of Treatment A over Treatment B.

If $\theta > 0$, Treatment A is more effective.

We wish to test the null hypothesis H_0: $\theta \le 0$ against $\theta > 0$ with

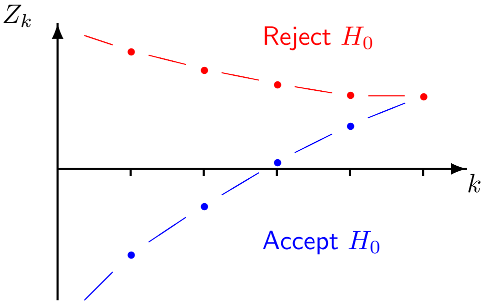
$$P_{\theta=0}\{\text{Reject } H_0\} = \alpha,$$

$$P_{\theta=\delta}\{\text{Reject } H_0\} = 1 - \beta.$$

In a group sequential trial, data are examined on a number of occasions to see if an early decision may be possible.

Group sequential tests

A typical boundary for a one-sided test, expressed in terms of standardised test statistics $Z_1, \ldots, Z_K$ has the form:



Crossing the upper boundary leads to early stopping for a positive outcome, rejecting H_0 in favour of $\theta > 0$.

Crossing the lower boundary implies stopping for "futility" with acceptance of H_0.

Benefits of group sequential testing

Earlier decisions

Group sequential testing can speed up the process to introduce an effective new treatment.

Fewer patients recruited

Expected sample sizes for group sequential designs are, typically, around 60 to 70% of the fixed sample size for a trial with the same type I error rate and power.

Stopping failing trials early

Early stopping "for futility" can release resources to continue the development of other promising treatments.

1.2. Joint distribution of parameter estimates

Reference: Ch. 11 of *Group Sequential Methods with Applications to Clinical Trials*, Jennison & Turnbull, 2000 (hereafter, JT).

Let $\widehat{\theta}_k$ denote the estimate of θ based on data at analysis k.

The information for θ at analysis k is

$$\mathcal{I}_k = \{\mathsf{Var}(\widehat{\theta}_k)\}^{-1}, \quad k = 1, \ldots, K.$$

Canonical joint distribution of $\widehat{\theta}_1, \ldots, \widehat{\theta}_K$

In many situations, $\widehat{\theta}_1, \ldots, \widehat{\theta}_K$ are approximately multivariate normal,

$$\widehat{\theta}_k \sim N(\theta, \mathcal{I}_k^{-1}), \quad k = 1, \ldots, K,$$

and

$$\mathsf{Cov}(\widehat{\theta}_{k_1}, \widehat{\theta}_{k_2}) = \mathsf{Var}(\widehat{\theta}_{k_2}) = \mathcal{I}_{k_2}^{-1} \quad \text{for } k_1 < k_2.$$

Sequential distribution theory

The joint distribution of $\widehat{\theta}_1, \ldots, \widehat{\theta}_K$ can be derived directly for:

θ a single normal mean,

$\theta = \mu_A - \mu_B,$ comparing two normal means.

The canonical distribution also applies when θ is a parameter in:

a general normal linear model,

a general model fitted by maximum likelihood (large sample theory).

Thus, theory supports general comparisons, including:

crossover studies,

analysis of longitudinal data,

comparisons adjusted for covariates.

Canonical joint distribution of $\widehat{\theta}_1, \ldots, \widehat{\theta}_K$

A single normal mean

Suppose X_1, X_2, $\ldots$ are independent $N(\theta, \sigma^2)$ responses.

For $n_1 < n_2$, define

$$\widehat{\theta}_1 = \frac{X_1 + \ldots + X_{n_1}}{n_1}, \quad \widehat{\theta}_2 = \frac{X_1 + \ldots + X_{n_1} + \ldots + X_{n_2}}{n_2}.$$

The joint distribution of $\widehat{\theta}_1$ and $\widehat{\theta}_2$ is bivariate normal.

Marginally

$$\widehat{\theta}_1 \sim N(\theta, \mathcal{I}_1{}^{-1}) \quad \text{and} \quad \widehat{\theta}_2 \sim N(\theta, \mathcal{I}_2{}^{-1}),$$

where

$$\mathcal{I}_1 = \frac{n_1}{\sigma^2} \quad \text{and} \quad \mathcal{I}_2 = \frac{n_2}{\sigma^2}.$$

Canonical joint distribution of $\widehat{\theta}_1, \ldots, \widehat{\theta}_K$

It remains to check the covariance:

$$
\begin{aligned}
\text{Cov}(\widehat{\theta}_1, \widehat{\theta}_2) &= \text{Cov}\left(\frac{X_1 + \ldots + X_{n_1}}{n_1}, \frac{X_1 + \ldots + X_{n_1} + \ldots + X_{n_2}}{n_2} \right) \\
&= \text{Cov}\left(\frac{X_1 + \ldots + X_{n_1}}{n_1}, \frac{X_1 + \ldots + X_{n_1}}{n_2} \right) \\
&= \frac{1}{n_1\, n_2} \text{Var}(X_1 + \ldots + X_{n_1}) \\
&= \frac{\sigma^2}{n_2} = \mathcal{I}_2^{-1} \\
&= \text{Var}(\widehat{\theta}_2).
\end{aligned}
$$

Canonical joint distribution of $\widehat{\theta}_1, \ldots, \widehat{\theta}_K$

A two-treatment comparison

Suppose observations on treatments A and B, respectively, are

$$X_{Ai} \sim N(\mu_A, \sigma^2) \quad \text{and} \quad X_{Bi} \sim N(\mu_B, \sigma^2),$$

and $\theta = \mu_A - \mu_B$.

At analysis k, with n_{Ak} observations on treatment A and n_{Bk} on treatment B,

$$\widehat{\theta}_k \;=\; \widehat{\mu}_{A,k} - \widehat{\mu}_{B,k} \;=\; \frac{1}{n_{Ak}} \sum_{i=1}^{n_{Ak}} X_{Ai} - \frac{1}{n_{Bk}} \sum_{i=1}^{n_{Bk}} X_{Bi}.$$

Exercise: Show that $\widehat{\theta}_1, \ldots, \widehat{\theta}_K$ have the canonical joint distribution, i.e., they are multivariate normal,

$$\widehat{\theta}_k \sim N(\theta, \mathcal{I}_k{}^{-1}), \quad \text{where} \quad \mathcal{I}_k = \{\sigma^2/n_{Ak} + \sigma^2/n_{Bk}\}^{-1}$$

and $\mathrm{Cov}(\widehat{\theta}_{k_1}, \widehat{\theta}_{k_2}) = \mathrm{Var}(\widehat{\theta}_{k_2})$ for $k_1 < k_2$.

Canonical joint distribution of $\widehat{\theta}_1, \ldots, \widehat{\theta}_K$

All efficient estimators have the canonical covariance

That is, if $\widehat{\theta}_1$ at analysis 1 and $\widehat{\theta}_2$ at analysis 2 are efficient, unbiased estimators of θ, then

$$\text{Cov}(\widehat{\theta}_1, \widehat{\theta}_2) = \text{Var}(\widehat{\theta}_2).$$

Proof:

Suppose $\text{Cov}(\widehat{\theta}_1, \widehat{\theta}_2) \neq \text{Var}(\widehat{\theta}_2)$, so $\text{Cov}(\widehat{\theta}_1 - \widehat{\theta}_2,\, \widehat{\theta}_2) \neq 0$.

Consider an unbiased estimator of the form $\widehat{\theta}_2^{\,*} = \widehat{\theta}_2 + \epsilon\,(\widehat{\theta}_1 - \widehat{\theta}_2)$.

For ϵ small and of the opposite sign to $\text{Cov}(\widehat{\theta}_1 - \widehat{\theta}_2,\, \widehat{\theta}_2)$,

$$\text{Var}(\widehat{\theta}_2^{\,*}) < \text{Var}(\widehat{\theta}_2),$$

contradicting the assumption that $\widehat{\theta}_2$ is an efficient estimator of θ.

Canonical joint distribution of z-statistics

In testing H_0: $\theta = 0$, the *standardised statistic* at analysis k is

$$Z_k = \frac{\widehat{\theta}_k}{\sqrt{\mathrm{Var}(\widehat{\theta}_k)}} = \widehat{\theta}_k \sqrt{\mathcal{I}_k}.$$

For these statistics,

$(Z_1, \ldots, Z_K)$ is multivariate normal,

$Z_k \sim N(\theta\sqrt{\mathcal{I}_k}, 1), \quad k = 1, \ldots, K$,

$\mathrm{Cov}(Z_{k_1}, Z_{k_2}) = \sqrt{\mathcal{I}_{k_1}/\mathcal{I}_{k_2}}$ for $k_1 < k_2$.

Canonical joint distribution of score statistics

The *score statistics*, $S_k = Z_k \sqrt{\mathcal{I}_k}$, are also multivariate normal with

$$S_k \sim N(\theta\, \mathcal{I}_k,\, \mathcal{I}_k), \quad k = 1, \ldots, K.$$

The score statistics possess the "independent increments" property,

$$\text{Cov}(S_k - S_{k-1},\, S_{k'} - S_{k'-1}) = 0 \quad \text{for } k \neq k'.$$

It can be helpful to know that the score statistics behave as Brownian motion with drift θ observed at times $\mathcal{I}_1, \ldots, \mathcal{I}_K$.

Survival data

The canonical joint distributions also arise for

a) estimates of a parameter in Cox's proportional hazards regression model,

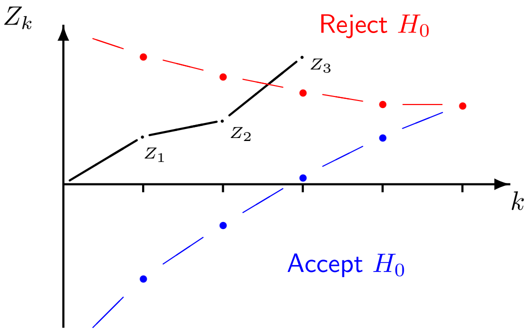
b) log-rank statistics for comparing two survival curves.

For survival data, observed information is roughly proportional to the number of failures.

The "error spending" approach can be used to define group sequential tests that can handle unpredictable and unevenly spaced information levels.

Reference: "Group-sequential analysis incorporating covariate information", Jennison & Turnbull (*JASA*, 1997).

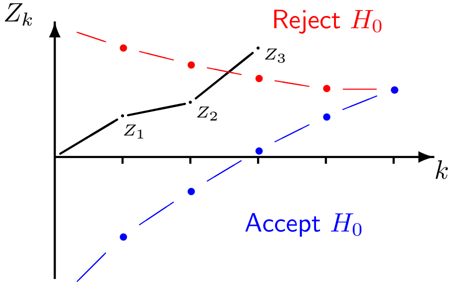
1.3. Computations for group sequential tests



In order to find $P_\theta\{\text{Reject } H_0\}$, etc., we need to calculate the probabilities of basic events such as

$$a_1 < Z_1 < b_1, \;\; a_2 < Z_2 < b_2, \;\; Z_3 > b_3.$$

Computations for group sequential tests



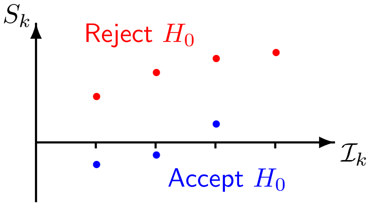
Probabilities such as $P_\theta\{a_1 < Z_1 < b_1,\ \ a_2 < Z_2 < b_2,\ \ Z_3 > b_3\}$ can be computed by repeated numerical integration (JT, Ch. 19).

Combining these probabilities yields type I error rate, power, expected sample size, etc., of a group sequential design.

Constants and group sizes can be chosen to define a test with a specific type I error probability and power.

One-sided tests: The Pampallona & Tsiatis family

To test H_0: $\theta \leq 0$ against the *one-sided* alternative $\theta > 0$ with type I error probability α and power $1 - \beta$ at $\theta = \delta$.



For the P & T test with parameter Δ, boundaries on the score statistic scale are

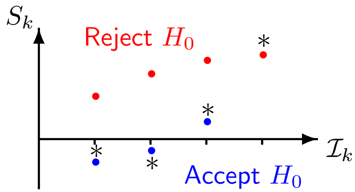
$$a_k = \mathcal{I}_k \, \delta - C_2 \, \mathcal{I}_k^{\Delta}, \quad b_k = C_1 \, \mathcal{I}_k^{\Delta}.$$

The computational methods described above can be used to find C_1, C_2 and $\mathcal{I}_K$ such that the test has the specified error rates.

Reference: Pampallona & Tsiatis (*JSPI*, 1994).

One-sided tests with a non-binding futility boundary

Regulators are not always convinced a trial monitoring committee will abide by the stopping boundary specified in the protocol.



The sample path shown above leads to rejection of H_0. Since such paths are not included in type I error calculations, the true type I error rate is under-estimated.

If a futility boundary is deemed to be *non-binding*, the type I error rate should be computed ignoring the futility boundary.

However, investigators will wish to know power and expected sample size when the futility boundary *is* obeyed.

1.4. Benefits of group sequential testing

In order to test H_0: $\theta \leq 0$ against $\theta > 0$ with type I error probability α and power $1-\beta$ at $\theta = \delta$, a fixed sample size study needs information

$$\mathcal{I}_{fix} = \frac{\{\Phi^{-1}(1-\alpha) + \Phi^{-1}(1-\beta)\}^2}{\delta^2},$$

where Φ is the standard normal cdf.

Information is (roughly) proportional to sample size in many clinical trial settings.

A group sequential test with K analyses will need to be able to continue to a maximum information level $\mathcal{I}_K$, greater than $\mathcal{I}_{fix}$.

On average, the sequential test can stop earlier than this and expected information on termination, $\mathbb{E}_\theta(\mathcal{I})$, will be considerably less than $\mathcal{I}_{fix}$, especially under extreme values of θ.

We call $R = \mathcal{I}_K / \mathcal{I}_{fix}$ the *inflation factor* of a group sequential test.

Optimal group sequential tests

We can seek a group sequential test that minimises expected information $\mathbb{E}_\theta(\mathcal{I})$ under certain values of the treatment effect, θ, with a given number of analyses K and inflation factor R.

Eales & Jennison (*Biometrika*, 1992) and Barber & Jennison (*Biometrika*, 2002) optimise designs for criteria of the form

$$\sum_i w_i \mathbb{E}_{\theta_i}(\mathcal{I}) \quad \text{or} \quad \int f(\theta)\, \mathbb{E}_\theta(\mathcal{I})\, d\theta,$$

where f is a normal density.

These optimised designs could be used in their own right.

They also serve as benchmarks for other methods which may have additional useful features.

Computing optimal group sequential tests

In optimising a group sequential test, we create a Bayes sequential decision problem, placing a prior on θ and defining costs for sampling and for making incorrect decisions.

Such a problem can be solved rapidly by dynamic programming.

We then search for the combination of prior and costs such that the solution to the (unconstrained) Bayes decision problem has the specified frequentist error rates α at $\theta = 0$ and β at $\theta = \delta$.

The resulting design solves both the Bayes decision problem and the original frequentist problem.

NB: Although the Bayes decision problem is introduced as a computational device, this derivation demonstrates that an efficient frequentist design should also be a good Bayesian procedure.

Benefits of group sequential testing

One-sided tests with binding futility boundaries, minimising $\{\mathbb{E}_0(\mathcal{I}) + \mathbb{E}_\delta(\mathcal{I})\}/2$ for K equally sized groups, $\alpha = 0.025$, $1 - \beta = 0.9$ and $\mathcal{I}_{max} = R\mathcal{I}_{fix}$.

Minimum values of $\{\mathbb{E}_0(\mathcal{I}) + \mathbb{E}_\delta(\mathcal{I})\}/2$, as a percentage of $\mathcal{I}_{fix}$

			R			*Minimum*
K	1.01	1.05	1.1	1.2	1.3	*over* R
2	80.8	74.7	**73.2**	73.7	75.8	73.0 at R=1.13
3	76.2	69.3	66.6	**65.1**	65.2	65.0 at R=1.23
5	72.2	65.2	62.2	59.8	**59.0**	58.8 at R=1.38
10	69.2	62.2	59.0	56.3	**55.1**	54.2 at R=1.6
20	67.8	60.6	57.5	54.6	**53.3**	51.7 at R=1.8

Note: $\mathbb{E}(\mathcal{I}) \searrow$ as $K \nearrow$ but with diminishing returns,

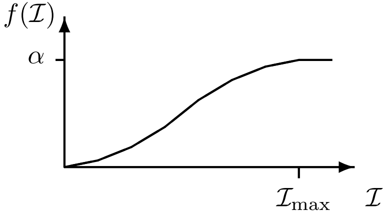
$\mathbb{E}(\mathcal{I}) \searrow$ as $R \nearrow$ up to a point.

1.5. Error spending tests

When the sequence $\mathcal{I}_1$, $\mathcal{I}_2$, $\ldots$ is unpredictable, a group sequential design must adapt to observed information levels.

Lan & DeMets (*Biometrika*, 1983) introduced "error spending" tests of H_0: $\theta = 0$ against $\theta \neq 0$.

Maximum information design with error spending function $f(\mathcal{I})$



The boundary at analysis k is set to give cumulative type I error probability $f(\mathcal{I}_k)$.

If $\mathcal{I}_{\max}$ is reached without rejecting H_0, then H_0, is accepted.

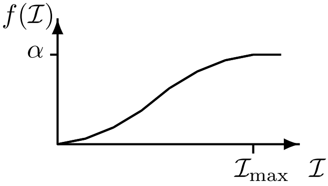
One-sided error spending tests

For a one-sided test of H_0: $\theta \leq 0$ against $\theta > 0$ with

Type I error probability α at $\theta = 0$,

Type II error probability β at $\theta = \delta$,

we need two error spending functions.



Type I error probability α is spent according to the function $f(\mathcal{I})$, and type II error probability β according to $g(\mathcal{I})$.

One-sided error-spending tests

Analysis 1:

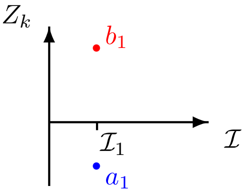
Observed information $\mathcal{I}_1$.

Reject H_0 if $Z_1 > b_1$, where

$$P_{\theta=0}\{Z_1 > b_1\} = f(\mathcal{I}_1).$$

Accept H_0 if $Z_1 < a_1$, where

$$P_{\theta=\delta}\{Z_1 < a_1\} = g(\mathcal{I}_1).$$



One-sided error-spending tests

Analysis 2: Observed information $\mathcal{I}_2$

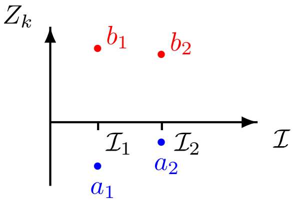
Reject H_0 if $Z_2 > b_2$, where

$$P_{\theta=0}\{a_1 < Z_1 < b_1, Z_2 > b_2\} = f(\mathcal{I}_2) - f(\mathcal{I}_1)$$

— *note that, for now, we assume the futility boundary is binding.*

Accept H_0 if $Z_2 < a_2$, where

$$P_{\theta=\delta}\{a_1 < Z_1 < b_1, Z_2 < a_2\} = g(\mathcal{I}_2) - g(\mathcal{I}_1).$$



One-sided error-spending tests

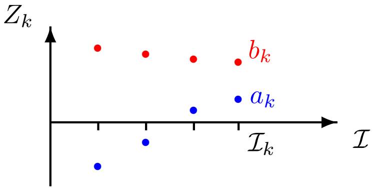
Analysis k: Observed information $\mathcal{I}_k$

Find a_k and b_k to satisfy

$$P_{\theta=0}\{a_1 < Z_1 < b_1, \ldots, a_{k-1} < Z_{k-1} < b_{k-1}, Z_k > b_k\} = f(\mathcal{I}_k) - f(\mathcal{I}_{k-1}),$$

and

$$P_{\theta=\delta}\{a_1 < Z_1 < b_1, \ldots, a_{k-1} < Z_{k-1} < b_{k-1}, Z_k < a_k\} = g(\mathcal{I}_k) - g(\mathcal{I}_{k-1}).$$

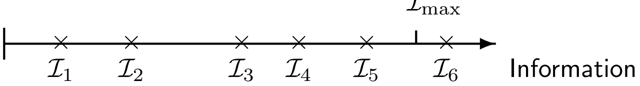


Remarks on error spending tests

1. Computation of (a_k, b_k) does **not** depend on future information levels, $\mathcal{I}_{k+1}$, $\mathcal{I}_{k+2}$, ...

2. A "maximum information design" continues until a boundary is crossed or an analysis with $\mathcal{I}_k \geq \mathcal{I}_{\max}$ is reached.

If necessary, patient accrual can be extended to reach $\mathcal{I}_{\max}$.



3. If a maximum of K analyses is specified, the study terminates at analysis K with $f(\mathcal{I}_K)$ defined to be α.

Then, a_K is chosen to give cumulative type I error probability α and we set $b_K = a_K$.

Remarks on error spending tests

4. The value of $\mathcal{I}_{\max}$ can be chosen so that boundaries converge at the final analysis when, say,

$$\mathcal{I}_k = (k/K)\,\mathcal{I}_{\max}, \quad k = 1, \ldots, K.$$

5. In a one-sided test with ρ-family error spending function, type I error probability is spent as

$$f(\mathcal{I}) = \alpha \, \min\{1, \, (\mathcal{I}/\mathcal{I}_{\max})^{\rho}\}$$

and type II error probability as

$$g(\mathcal{I}) = \beta \, \min\{1, \, (\mathcal{I}/\mathcal{I}_{\max})^{\rho}\}.$$

The value of ρ determines the inflation factor $R = \mathcal{I}_{\max}/\mathcal{I}_{fix}$.

Barber & Jennison (*Biometrika*, 2002) show the ρ-family provides tests with excellent efficiency for a given number of analyses K and inflation factor R.

One-sided error-spending tests: non-binding futility

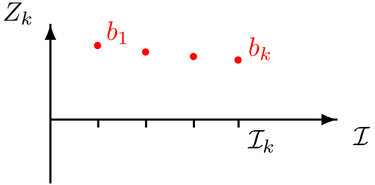
If the futility boundary is treated as non-binding, computation of the efficacy boundary only involves the type I error spending function $f(\mathcal{I})$.

Boundary values, b_1, b_2, $\ldots$, are calculated as the trial proceeds.

Analysis k: Observed information $\mathcal{I}_k$

Reject H_0 if $Z_k > b_k$, where

$$P_{\theta=0}\{Z_1 < b_1, \ldots, Z_{k-1} < b_{k-1}, Z_k > b_k\} = f(\mathcal{I}_k) - f(\mathcal{I}_{k-1}).$$



One-sided error-spending tests: non-binding futility

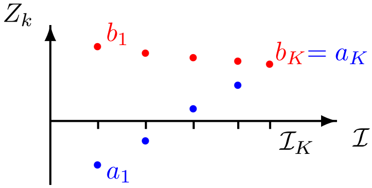
A futility boundary can be added through a type II error spending function $g(\mathcal{I})$.

For $k = 1, \ldots, K-1$*:*

At analysis k with observed information $\mathcal{I}_k$, set a_k to satisfy

$$P_{\theta=\delta}\{a_1 < Z_1 < b_1, \ldots, a_{k-1} < Z_{k-1} < b_{k-1}, Z_k < a_k\} = g(\mathcal{I}_k) - g(\mathcal{I}_{k-1}).$$

For $k = K$*:* Set $a_K = b_K$.



Example: Oropharynx Clinical Trial Data

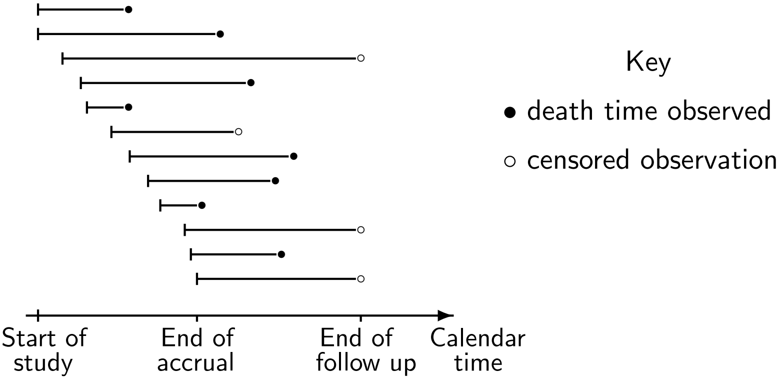
Survival of patients on experimental Treatment A and standard Treatment B.

Analysis		Number entered		Number of deaths	
k	Date	Trt A	Trt B	Trt A	Trt B
1	12/69	38	45	13	14
2	12/70	56	70	30	28
3	12/71	81	93	44	47
4	12/72	95	100	63	66
5	12/73	95	100	69	73

From Kalbfleisch & Prentice (2002) *The Statistical Analysis of Failure Time Data, 2nd edition*, Appendix A, Data Set II.

See also JT, Ch. 13.

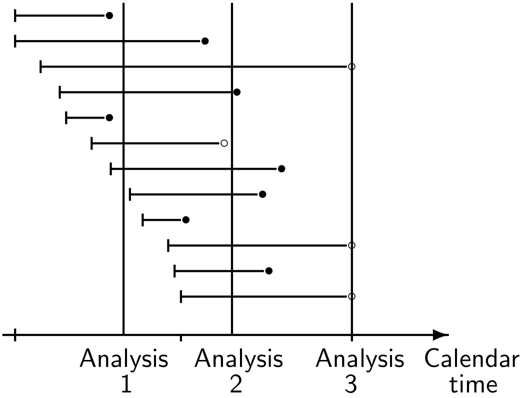
Accrual and follow up in a survival study



Subjects are randomised to a treatment as they enter the study.

Survival is measured from entry to the study.

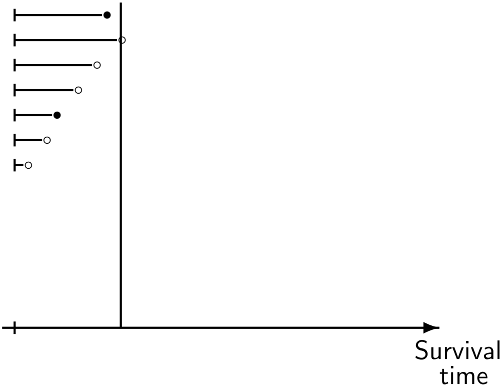
Interim analyses



At an interim analysis, subjects are censored if they are still alive.

Information on such patients continues to accrue at later analyses.

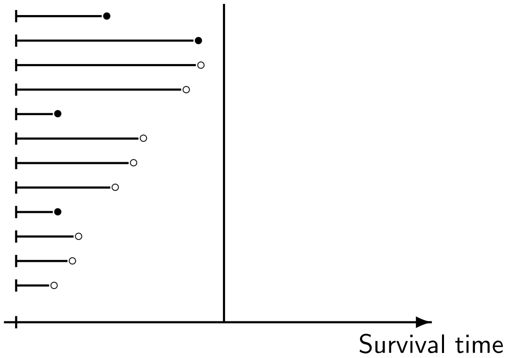
Interim analysis 1



We analyse data on survival from time of randomisation.

Survival times start at zero and “analysis time” censoring occurs for subjects surviving past this first analysis.

Interim analysis 2



At interim analysis 2, there is further follow-up of subjects who were censored at analysis 1.

In addition, there is initial information on the survival times of subjects entering the trial since analysis 1.

The logrank statistic

At stage k, the observed number of deaths is d_k.

Elapsed times between entry to the study and these deaths are

$$\tau_{1,k} < \tau_{2,k} < \ldots < \tau_{d_k,k} \quad \text{(assuming no ties).}$$

Define variables at analysis k

$r_{iA,k}$ and $r_{iB,k}$	Numbers at risk on Trts A and B at $\tau_{i,k}-$
$r_{ik} = r_{iA,k} + r_{iB,k}$	Total number at risk at $\tau_{i,k}-$
O_k	Observed number of deaths on Trt B
$E_k = \sum_{i=1}^{d_k} r_{iB,k}/r_{ik}$	"Expected" number of deaths on Trt B
$V_k = \sum_1^{d_k} r_{iA,k} r_{iB,k}/r_{ik}^2$	"Variance" of O_k
$Z_k = (O_k - E_k)/\sqrt{V_k}$	Standardised logrank statistic

Proportional hazards model

Assume hazard rates h_A on Treatment A and h_B on Treatment B are related by

$$h_B(t) = \lambda\, h_A(t).$$

The log hazard ratio is $\theta = \ln(\lambda)$.

Then, with $\mathcal{I}_k = V_k$, we have approximately

$$Z_k \sim N(\theta\sqrt{\mathcal{I}_k},\, 1), \quad k = 1, \ldots, K,$$

$$\text{Cov}(Z_{k_1}, Z_{k_2}) = \sqrt{(\mathcal{I}_{k_1}/\mathcal{I}_{k_2})} \quad \text{for } k_1 < k_2.$$

In addition, $(Z_1, \ldots, Z_K)$ is approximately multivariate normal — so the statistics $Z_1, \ldots, Z_K$ follow the canonical joint distribution.

The related score statistic at analysis k is $Z_k\sqrt{\mathcal{I}_k}$ and this has variance $V_k = \mathcal{I}_k$.

Proportional hazards model

Since $Z_k \sim N(\theta\sqrt{\mathcal{I}_k},\, 1)$, we can estimate θ at analysis k by

$$\widehat{\theta}_k = \frac{Z_k}{\sqrt{\mathcal{I}_k}}.$$

It follows that

$$\widehat{\theta}_k \; \sim N(\theta,\, \mathcal{I}_k^{-1}) \quad \text{approximately.}$$

Recall that

$$V_k \; = \; \sum_{i=1}^{d_k} \frac{r_{iA,k}\, r_{iB,k}}{(r_{iA,k} + r_{iB,k})^2}.$$

If equal numbers are randomised to treatments A and B and $\lambda \approx 1$, we can expect $r_{iA,k} \approx r_{iB,k}$ for each k, and so

$$\mathcal{I}_k = V_k \approx d_k/4.$$

Design of the Oropharynx trial

Suppose we wish to create a one-sided test of H_0: $\theta \leq 0$ vs $\theta > 0$.

Note $\theta > 0 \;\Rightarrow\; \lambda > 1,$ i.e., Treatment A is better.

We require:

Type I error probability $\alpha = 0.025,$

Power $1 - \beta = 0.8$ at $\theta = 0.5,$ i.e., at $\lambda = 1.65.$

Information needed for a fixed sample study is

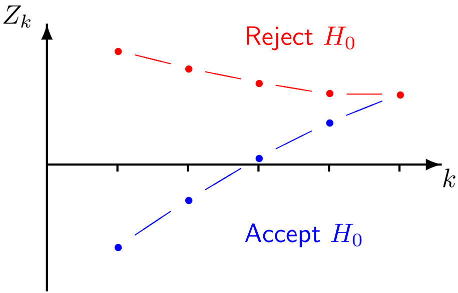
$$\mathcal{I}_f \;=\; \frac{\{\Phi^{-1}(1-\alpha) + \Phi^{-1}(1-\beta)\}^2}{0.5^2} \;=\; 31.40.$$

Under the approximation $\mathcal{I} \approx d/4$, the total number of failures to be observed is

$$d_f \;=\; 4\mathcal{I}_f \;\approx\; 126.$$

Design of the Oropharynx trial

For a one-sided test with up to 5 analyses, we could use a standard design created for equally spaced information levels.



However, increments in information between analyses are unpredictable.

So, an error spending design is a natural choice.

A one-sided, error spending design

Specification:

One-sided test of H_0: $\theta \leq 0$ vs $\theta > 0$,

Type I error probability $\alpha = 0.025$,

Power $1 - \beta = 0.8$ at $\theta = \ln(\lambda) = 0.5$,

Binding futility boundary.

When designing, assume $K = 5$ equally spaced information levels.

Use a power-family test with $\rho = 2$ to spend error $\propto (\mathcal{I}/\mathcal{I}_{\max})^2$.

Information for a fixed sample test has to be inflated by $R = 1.098$.

So, we require $\mathcal{I}_{\max} = 1.098 \times 31.40 = 34.48$, which needs a total of $4 \times 34.48 \approx 138$ observed deaths.

A one-sided, error spending design

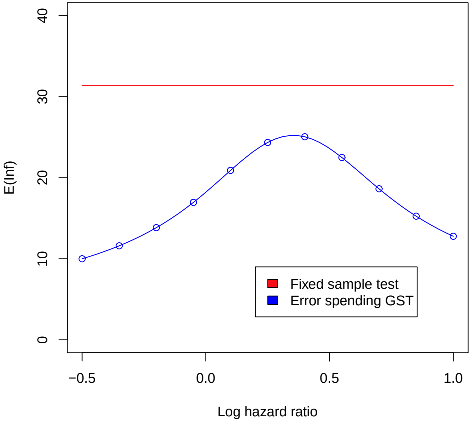
Suppose that, as assumed when planning the trial, information levels are equally spaced up to $\mathcal{I}_5 = \mathcal{I}_{\max} = 34.48$.

Then, we would have the following boundary values $(a_1, b_1), \ldots, (a_5, b_5)$ for the standardised logrank statistics $Z_1, \ldots, Z_5$.

k	$\mathcal{I}_k$	a_k	b_k
1	6.90	−1.10	3.09
2	13.79	−0.05	2.71
3	20.69	0.72	2.47
4	27.58	1.39	2.28
5	34.48	2.06	2.06

A one-sided, error spending design

If information levels $\mathcal{I}_k = (k/5)\,34.48,\ k = 1, \ldots, 5,$ are observed, the expected information on termination is the following function of the log hazard ratio, θ.



Summary data and critical values for the Oropharynx trial

In reality, we construct error spending boundaries using the *observed* information levels.

This gives the following boundary values $(a_1, b_1), \dots, (a_5, b_5)$ for the standardised logrank statistics $Z_1, \dots, Z_5$.

Analysis k	*Number entered*	*Number of deaths*	$\mathcal{I}_k$	a_k	b_k	Z_k
1	83	27	5.43	−1.41	3.23	−1.04
2	126	58	12.58	−0.21	2.76	−1.00
3	174	91	21.11	0.78	2.44	−1.21
4	195	129	30.55	1.68	2.16	−0.73
5	195	142	33.28	2.14	2.14	−0.87

The trial would have terminated at analysis 2 to accept H_0.

An error spending test with a non-binding futility boundary

If a **non-binding** futility boundary is used, the required maximum information level is a little higher at 35.58.

Applying this design to the observed information levels gives:

Analysis k	*Number entered*	*Number of deaths*	$\mathcal{I}_k$	a_k	b_k	Z_k
1	83	27	5.43	−1.44	3.25	−1.04
2	126	58	12.58	−0.23	2.78	−1.00
3	174	91	21.11	0.75	2.46	−1.21
4	195	129	30.55	1.64	2.20	−0.73
5	195	142	33.28	2.09	2.09	−0.87

Again, the trial terminates at analysis 2 with acceptance of H_0.

Covariate adjustment in the Oropharynx trial

Covariate information was recorded for subjects: *Institution (6), Gender, Initial condition, T-staging, N-staging, Tumor site (3).*

A proportional hazards regression model includes

Strata $l = 1, \ldots, 6$ for the six participating institutions,

Treatment effect β_1,

Coefficients $\beta_2, \ldots, \beta_5$ for Gender and the continuous variables Initial condition, T-staging and N-staging,

Coefficients β_6 and β_7 for the categorical variable Tumor site.

Modelling the hazard rate for patient i as

$$h_{il}(t) = h_{0l}(t)\, e^{\{\beta_1 I(\text{Patient } i \text{ on Trt B}) + \Sigma_{j=2}^{7} x_{ij}\beta_j\}},$$

the objective is to test H_0: $\beta_1 = 0$ against $\beta_1 > 0$.

Covariate adjustment in the Oropharynx trial

Standard software for Cox regression can provide an estimate of the parameter vector, β, and its estimated variance.

We are interested in the treatment effect β_1.

At stage k we have

$$\widehat{\beta}_1^{(k)}$$

$$v_k = \widehat{\mathsf{Var}}\left(\widehat{\beta}_1^{(k)}\right)$$

$$\mathcal{I}_k = v_k^{-1}$$

$$Z_k = \widehat{\beta}_1^{(k)}/\sqrt{v_k}.$$

Theory tells us: The standardised statistics $Z_1, \ldots, Z_5$ have, approximately, the canonical joint distribution.

Covariate-adjusted analysis of the Oropharynx trial

Constructing the error spending test with a **non-binding** futility boundary gives critical values $(a_1, b_1), \ldots, (a_5, b_5)$ for $Z_1, \ldots, Z_5$.

k	$\mathcal{I}_k$	a_k	b_k	$\widehat{\beta}_1^{(k)}$	Z_k
1	4.11	−1.77	3.40	−0.79	−1.60
2	10.89	−0.47	2.87	−0.14	−0.45
3	19.23	0.55	2.52	−0.08	−0.33
4	28.10	1.41	2.27	0.04	0.20
5	30.96	2.27	2.27	0.01	0.04

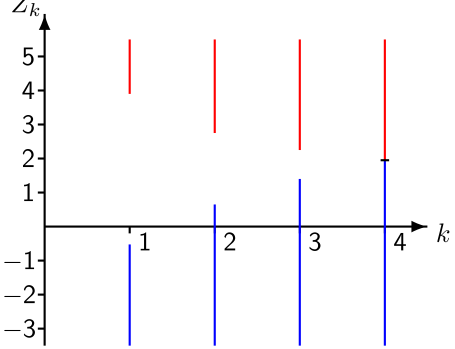
Under this stopping rule, the study would have continued — just — at analysis 2 and stopped to accept H_0 at analysis 3.

Note that β_1 is the log hazard ratio after covariate adjustment. For $\beta_1 > 0$, we should expect $\beta_1 > \lambda$ where λ is the log hazard ratio in a model without covariates.

1.7. Analysis on termination of a group sequential test

Our sample space is all possible pairs (k, Z_k) on termination.

Sample space for a Pampallona & Tsiatis with $\Delta = 0$, $K = 4$ analyses, $\alpha = 0.025$ and power $1 - \beta = 0.8$ at $\theta = 1$:



Frequentist inference involves probabilities on this *sample space*.

The need for special methods

Suppose our 4-stage study with a Pampallona & Tsiatis boundary ends at stage 3 with $Z_3 = 2.6$.

It may be tempting to quote a 1-sided P-value of

$$P\{N(0,1) > 2.60\} = 0.0047.$$

But, using this definition, we would also get a P-value ≤ 0.0047 by

stopping at stage 1 with $Z_1 > 3.90$,

stopping at stage 2 with $Z_2 > 2.76$,

stopping at stage 3 with $Z_3 > 2.60$,

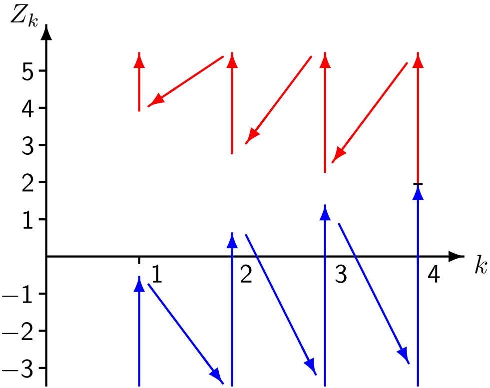
stopping at stage 4 with $Z_4 > 2.60$,

and the total probability under $\theta = 0$ of "$P \leq 0.0047$" is 0.0076.

So, this "P-value" does *not* have the null distribution $U(0,1)$.

Analysis on termination of a group sequential test (GST)

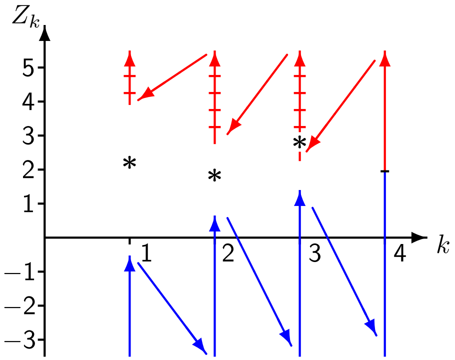
For proper frequentist inference, we first order the sample space.



Then, we define P-values and confidence intervals with respect to this ordering.

(i) A P-value on termination

The *P-value* for H_0: $\theta = 0$ is the probability under H_0 of seeing an outcome as extreme as that observed.



On stopping with $Z_3 = 2.60$, the 1-sided P-value for H_0: $\theta \leq 0$ is

$P_{\theta=0}\{$Stop with $Z_1 \geq 3.90$ or $Z_2 \geq 2.76$ or $Z_3 \geq 2.60\} = 0.0063.$

A P-value on termination

With the above definition, based on a specific ordering of the sample space:

The P-value has a $U(0,1)$ distribution under H_0.

If the group sequential test has one-sided type I error probability α, the P-value is $\leq \alpha$ precisely when the test stops with rejection of H_0,

i.e., in the part of the sample space coloured red in the previous slide.

The P-value will tend to take low values when θ is large and positive.

(ii) A confidence interval on termination

First, we specify an ordering of the GST's sample space.

Suppose the trial terminates at analysis k^* with $Z_{k^*} = Z^*$.

We define the $100(1-2\alpha)\%$ confidence interval for θ to be the

set of values θ for which the observed (k^*, Z^*) is in the middle $(1-2\alpha)$ of the probability distribution under θ.

This is the interval $(\theta_1,\, \theta_2)$ where

$$P_{\theta=\theta_1}\{\text{An outcome above } (k^*, Z^*)\} = \alpha$$

and

$$P_{\theta=\theta_2}\{\text{An outcome below } (k^*, Z^*)\} = \alpha.$$

There is a duality between this $100(1-2\alpha)\%$ confidence interval and the family of level 2α, two-sided tests of hypotheses H: $\theta = \tilde{\theta}$.

A confidence interval on termination

Example:

Suppose the trial stops at analysis 3 with $Z_3 = 2.6$.

Using our specified ordering, the 95% confidence interval for θ is

$$(0.22,\ 1.77)$$

In contrast:

The "naive" fixed sample CI would be $(0.25,\ 1.78)$.

However, it is not appropriate to use this fixed sample interval as this fails to take account of the sequential stopping rule.

Consequently, the coverage probability of this naive, fixed sample interval is *not* $1 - 2\alpha$.

Consistency of hypothesis testing and CI on termination

Suppose a group sequential trial is run to test H_0: $\theta \leq 0$ vs $\theta > 0$ with one-sided type I error probability α.

Then, a $1 - 2\alpha$, equal-tailed confidence interval on termination should lie completely above $\theta = 0$ if and only if H_0 is rejected.

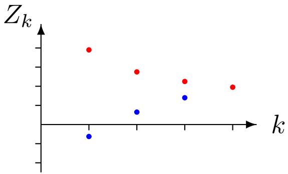
This happens automatically if outcomes for which we reject H_0 are at the top end of the sample space ordering — and any sensible ordering does this.

Why the naive approach does not work

A naive $1 - 2\alpha$ level CI on termination lies completely above $\theta = 0$ if an *unadjusted* α level, one-sided significance test rejects H_0.

Since there are multiple analyses, the probability of such an outcome is liable to be greater than the desired level α.

(iii) Estimating θ after a group sequential test



In a two-treatment comparison, the maximum likelihood estimate (MLE) of θ on termination of the trial at analysis k is

$$\widehat{\theta}_M \;=\; \bar{X}_{Ak} - \bar{X}_{Bk}.$$

For large, positive values of θ:

high values of $\widehat{\theta}$ lead to early stopping,

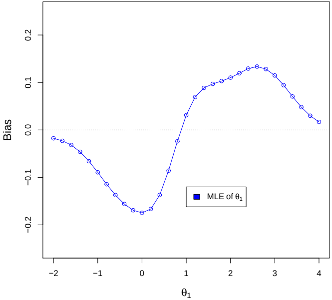
lower values of $\widehat{\theta}$ result in more observations, so $\widehat{\theta}$ can increase.

Thus, the MLE is biased with $E_\theta(\widehat{\theta}_M) > \theta$ for high values of θ.
Similarly, $E_\theta(\widehat{\theta}_M) < \theta$ for low values of θ.

Bias of the MLE of θ after a Pampallona & Tsiatis test

Consider the Pampallona & Tsiatis GST with $\Delta = 0$, $K = 4$ analyses, $\alpha = 0.025$ and power $1 - \beta = 0.8$ at $\theta = 1$

The bias of the MLE can be calculated as a function of the true effect size, θ.



The bias of the MLE is around 0.1 at values of θ just above 1.

Correcting the bias of the MLE

Denote the bias function of the MLE by

$$b(\theta) \,=\, E_\theta(\widehat{\theta}_M) - \theta.$$

Whitehead (*Biometrika*, 1986) suggested correcting the MLE by subtracting an estimate of its bias.

Although the true θ is unknown, the bias of the MLE can be estimated by $b(\widehat{\theta}_M)$.

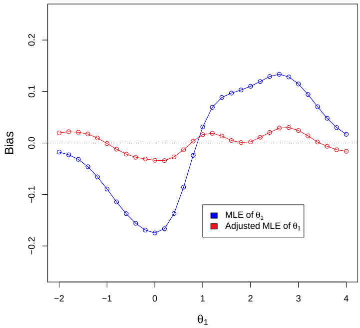
The adjusted estimator is then

$$\widehat{\theta}_{adj} \,=\, \widehat{\theta}_M - b(\widehat{\theta}_M).$$

Bias of the MLE of θ after a Pampallona & Tsiatis test

Simulation results show that Whitehead's adjusted estimator has much smaller bias than the MLE on which it is based.

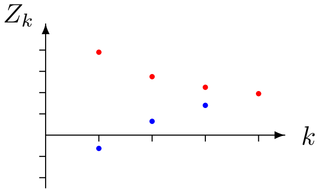
For our example:



The adjustment almost completely removes the bias in the MLE.

(iv) Estimation for a secondary endpoint after a GST

Stopping boundary for the primary endpoint



Denote the treatment effect on the primary endpoint by θ_1.

Suppose the trial stops and rejects H_0: $\theta_1 \leq 0$ in favour of $\theta_1 > 0$.

On stopping, data on a secondary endpoint are analysed to estimate the treatment effect, θ_2, on this endpoint.

For an individual, primary and secondary responses are correlated.

The group sequential design leads to bias in the MLE $\hat{\theta}_1$ — and the correlated responses imply that bias is passed on to the MLE $\hat{\theta}_2$.

Estimation for a secondary endpoint after a GST

Suppose an individual's responses are bivariate normal with correlation ρ.

For a patient on Treatment A,

$$\begin{array}{ll} \text{Primary endpoint} & X_1 \sim N(\mu_{A1},\, \sigma_1^2), \\ \text{Secondary endpoint} & X_2 \sim N(\mu_{A2},\, \sigma_2^2). \end{array}$$

Similarly, for a patient on Treatment B,

$$\begin{array}{ll} \text{Primary endpoint} & X_1 \sim N(\mu_{B1},\, \sigma_1^2), \\ \text{Secondary endpoint} & X_2 \sim N(\mu_{B2},\, \sigma_2^2). \end{array}$$

The primary treatment effect is

$$\theta_1 = \mu_{A1} - \mu_{B1}$$

and the secondary treatment effect is

$$\theta_2 = \mu_{A2} - \mu_{B2}.$$

Estimation for a secondary endpoint after a GST

Consider a group sequential design where the bias in the MLE $\hat{\theta}_1$ is

$$b_1(\theta) \;=\; E_\theta(\hat{\theta}_1) - \theta_1$$

when the true treatment effects are $\theta = (\theta_1, \theta_2)$.

Note that $E_\theta(\hat{\theta}_1)$ depends on θ_1 and not on θ_2.

Whitehead (*Biometrics*, 1986) shows that the MLE $\hat{\theta}_2$ has bias

$$b_2(\theta) \;=\; E_\theta(\hat{\theta}_2) - \theta_2 \;=\; \rho\,\sqrt{(\sigma_2^2/\sigma_1^2)}\;b_1(\theta)$$

when the true treatment effects are $\theta = (\theta_1, \theta_2)$.

Note that this bias depends on θ_1 — and not on θ_2.

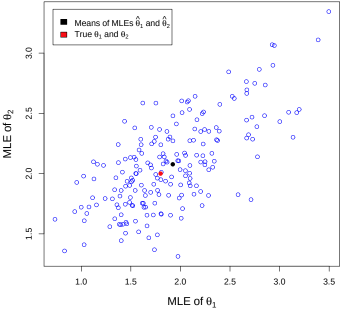
As for the primary endpoint, we can adjust the MLE, $\hat{\theta}_2$, by subtracting an estimate of its bias, $(\rho\,\sigma_2/\sigma_1)\,b_1(\hat{\theta})$.

Estimation for a secondary endpoint: Example

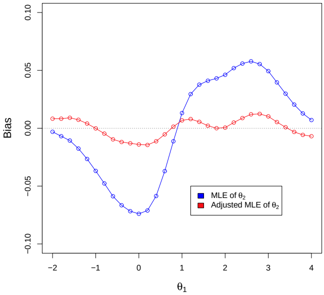
Suppose a trial tests its primary endpoint, using a Pampallona & Tsiatis GST with $\Delta = 0$, $\alpha = 0.025$ and power 0.8 at $\theta_1 = 1$.

Responses are bivariate normal, $\rho = 0.6$ and $\sigma_1^2/\sigma_2^2 = 2$.

The plot, for the case $\theta_1 = 1.8$ and $\theta_2 = 2$, shows the correlation between the MLEs, $\hat{\theta}_1$ and $\hat{\theta}_2$, on termination of the GST.



Estimation for a secondary endpoint: Example



We see that the bias in the MLE $\hat{\theta}_2$ is largely eliminated in the adjusted estimator

$$\hat{\theta}_2 - \rho \sqrt{\frac{\sigma_2^2}{\sigma_1^2}}\, b_1(\hat{\theta}).$$

(v) Testing a secondary endpoint after a GST

In a trial of two treatments, A and B, a group sequential test is carried out on the primary endpoint, which has treatment effect θ_1.

Suppose H_1: $\theta_1 \leq 0$ is rejected in favour of $\theta_1 > 0$.

The investigators wish to test whether Treatment A is also superior for a secondary endpoint, with treatment effect denoted by θ_2.

Some familiarity with "gatekeeping" procedures for testing multiple hypotheses suggests it may be legitimate to pass on the type I error $\alpha = 0.025$ to a second hypothesis test.

As this test will be only conducted once, the investigators plan to carry out a fixed sample size, level α test of H_2: $\theta_2 \leq 0$ vs $\theta_2 > 0$ using the available data on the secondary endpoint.

Is this approach to testing the two endpoints valid?

Testing a secondary endpoint: Example

Suppose the primary endpoint is tested using a Pampallona & Tsiatis group sequential design with shape parameter $\Delta = 0$.

There are 4 analyses, type I error probability is $\alpha = 0.025$ and power is 0.8 at $\theta_1 = 1$.

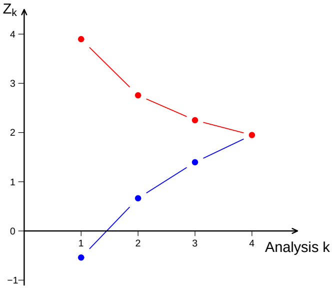
This test has upper boundary:

$$Z_k \;=\; 3.90/\sqrt{k}$$

and lower boundary

$$Z_k \;=\; 1.48\sqrt{k} - 2.02/\sqrt{k},$$

where $k = 1, \ldots, 4$.

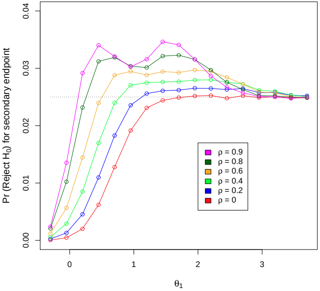


If the upper boundary is crossed, the secondary endpoint is tested in a level α, fixed sample size test, using current data.

Testing a secondary endpoint: Example

The plot shows the probability of rejecting H_2: $\theta_2 \leq 0$, under $\theta_2 = 0$, when the secondary endpoint is tested as described above.

The two endpoints have correlation ρ. For modest values of ρ, the type I error rate for testing H_2 exceeds the nominal 0.025.



Hung, Wang and O'Neill (*J. Biopharm. Statist.*, 2007) noted that this approach to testing a secondary endpoint is not valid.

So, how should the secondary endpoint be tested?

Recapitulation: Group sequential tests

- It is natural to monitor clinical trials with a view to possible early stopping.
- Distribution theory and computational methods support a variety of group sequential designs that control the type I error rate.
- These designs can be optimised for a given objective.
- Error spending designs offer efficient, flexible monitoring of a variety of response types, including survival data.
- Inference on termination can provide P-values, confidence intervals and approximately unbiased point estimates.
- In order to conduct **multiple** hypothesis tests within a group sequential design with proper control of the type I error rate, **we shall need additional methodology.**

Part 2. Multiple testing procedures

2.1 Multiple testing: The familywise error rate

We have just seen an example with one primary and one secondary endpoint.

More generally, a clinical trial may involve

Co-primary endpoints

Positive outcomes required for at least one endpoint

Positive outcomes required on all endpoints

Secondary endpoints, tertiary endpoints, ...

The trial may have

Multiple treatments,

Pre-defined sub-populations of patients.

If the trial is group sequential, each hypothesis may be tested on several occasions.

The familywise error rate

Suppose we have h null hypotheses, H_i: $\theta_i \leq 0$ for $i = 1, \ldots, h$.
After our analysis, we accept or reject each of these h hypotheses.

A testing procedure's **familywise error rate** under a set of values $\theta = (\theta_1, \ldots, \theta_h)$ is

$$P_\theta\{\text{Reject } H_i \text{ for some } i \text{ with } \theta_i \leq 0\}$$

$$= \; P_\theta\{\text{Reject at least one true } H_i\}.$$

The familywise error rate is controlled **strongly** at level α if this error rate is at most α for all possible combinations of θ_i values.

Then

$$P_\theta\{\text{Reject any true } H_i\} \; \leq \; \alpha \quad \text{for all } (\theta_1, \ldots, \theta_h).$$

2.2. Bonferroni adjustment (Carlo Bonferroni, 1892–1960)

Suppose we test h null hypotheses, each at significance level α/h.

If θ is such that all h null hypotheses are true,

$$P_\theta\{\text{Reject at least one of } H_1, \ldots, H_h\}$$

$$\leq\ P_\theta\{\text{Reject } H_1\} + \ldots + P_\theta\{\text{Reject } H_h\}\ \leq\ h\,\frac{\alpha}{h}\ =\ \alpha.$$

If θ is such that only some of the h null hypotheses are true,

$$P_\theta\{\text{Reject at least one true } H_i\}\ <\ \alpha.$$

So we have **strong control** of the **familywise error rate**.

We start by considering applications in fixed sample size study designs ...

Example: A Bonferroni test with co-primary endpoints

A trial compares a new treatment against control with respect to:

Endpoint 1, Core MACE (*Major Adverse Cardiac Event* — CV-related death, nonfatal stroke, or nonfatal MI)

Endpoint 2, Expanded MACE (Core MACE plus hospitalization for unstable angina or coronary revascularization).

Type I error probability $\alpha = 0.025$ is divided between the endpoints.

With Z-statistics Z_1 and Z_2 for endpoints 1 and 2,

An effect on Core MACE is declared if

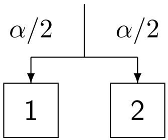
$$Z_1 > \Phi^{-1}(1 - \alpha/2) = 2.24,$$

An effect on Expanded MACE is declared if

$$Z_2 > \Phi^{-1}(1 - \alpha/2) = 2.24.$$

Example: Co-primary endpoints

This Bonferroni procedure can be represented graphically as:



There is a positive correlation between the two tests, due to the common aspects of the two endpoints.

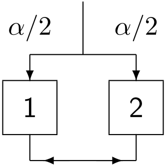
Hence, familywise type I error is protected conservatively.

Power when H_1 and H_2 are false can be increased by "recycling" type I error after one or other hypothesis is rejected.

2.3. Recycling type I error probability

The Holm procedure is a version of the Bonferroni procedure that "recycles" error probability after rejecting H_1 or H_2.

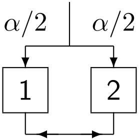
This method can be represented as:



If H_1 is rejected at level $\alpha/2$, we pass that error probability to H_2 and test this hypothesis at level α.

If H_2 is rejected at level $\alpha/2$, we pass that error probability to H_1 and test this hypothesis at level α.

Proof that FWER is protected



If H_1 and H_2 are both true,

$$\begin{aligned} \text{FWER} &= P_\theta\{\text{Reject } H_1 \text{ or } H_2\} \\ &\leq P_\theta\{Z_1 > \Phi^{-1}(1-\alpha/2)\} + P_\theta\{Z_2 > \Phi^{-1}(1-\alpha/2)\} \\ &\leq \alpha/2 + \alpha/2 \; = \; \alpha. \end{aligned}$$

If H_1 is true and H_2 is false,

$$\text{FWER} = P_\theta\{\text{Reject } H_1\} \leq P_\theta\{Z_1 > \Phi^{-1}(1-\alpha)\} \leq \alpha.$$

H_2 is true and H_1 false: Similar to H_1 true and H_2 false.

H_1 and H_2 both false: A type I error cannot be made.

2.4. Example: Primary and secondary endpoints

A hierarchical testing or "gatekeeping" procedure

Consider a trial where

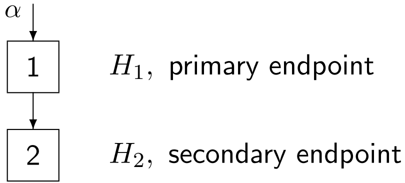
The null hypothesis H_1 concerns the primary endpoint,

The null hypothesis H_2 relates to a secondary endpoint,

and H_2 will only be tested if H_1 has already been rejected.

First, we test H_1 at significance level α.

If H_1 is rejected, we continue and test H_2 at significance level α.



Proof that FWER is protected

$$\alpha \downarrow \boxed{1} \downarrow \boxed{2}$$

$\boxed{1}$ H_1, primary endpoint

$\boxed{2}$ H_2, secondary endpoint

Suppose H_1 is true.

A family-wise error occurs if H_1 is rejected (whether or not H_2 is also rejected). So

$$\text{FWER} \;=\; P_\theta\{\text{Reject } H_1\} \;=\; P_\theta\{Z_1 > \Phi^{-1}(1-\alpha)\} \;\leq\; \alpha.$$

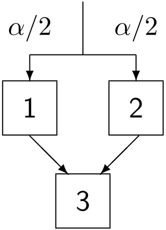
If H_1 is false and H_2 is true,

$$\begin{aligned} \text{FWER} \;&=\; P_\theta\{\text{Reject } H_1 \text{ and then reject } H_2\} \\ &\leq\; P_\theta\{Z_2 > \Phi^{-1}(1-\alpha)\} \;\leq\; \alpha. \end{aligned}$$

If H_1 and H_2 are both false, a type I error cannot be made.

Example: Testing co-primary and secondary endpoints

The figure below represents a testing procedure that starts with a Bonferroni test of H_1 and H_2.



H_1, H_2: co-primary endpoints

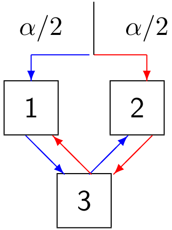
H_3: secondary endpoint

Then, if either H_1 or H_2 is rejected, the associated type I error is passed on to the test of H_3.

We can prove there is strong control of FWER at level α by considering all combinations of H_1, H_2 and H_3 being True or False.

Testing co-primary and secondary endpoints

We can add more "recycling" to the previous testing procedure.



H_1, H_2: co-primary endpoints

H_3: secondary endpoint

The additional lines in the graph indicate that

If $P_1 \leq \alpha/2$ and $P_3 \leq \alpha/2$, then H_2 is tested at level α,

If $P_2 \leq \alpha/2$ and $P_3 \leq \alpha/2$, then H_1 is tested at level α.

Testing co-primary and secondary endpoints

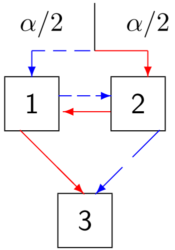
We may prefer to gain maximum power for tests of co-primary endpoints before testing a secondary endpoint.

To do this, we recycle type I error probability between H_1 and H_2 before allocating any error probability to H_3.

A graphical representation is:

H_1, H_2: co-primary endpoints

H_3: secondary endpoint



Half of the type I error probability is cycled through H_1, H_2 and on to H_3.

The other half is cycled through H_2, H_1 and on to H_3.

2.5. Graphical representation of multiple testing procedures

As we add more options, and get more creative, we can produce some quite complex procedures.

Two papers, published simultaneously, describe an elegant way to describe complex multiple testing procedures.

> *"A recycling framework for the construction of Bonferroni-based multiple tests" by Burman, Sonesson & Guilbaud,* Statistics in Medicine, *2009.*

> *"A graphical approach to sequentially rejective multiple test procedures" by Bretz, Maurer, Brannath & Posch,* Statistics in Medicine, *2009.*

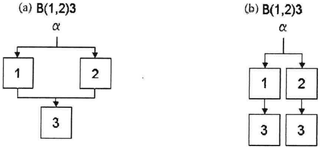
These procedures are closed testing procedures in which the tests of intersection hypotheses are weighted Bonferroni tests.

It is implicit in their method of construction that these procedures provide strong control of the FWER.

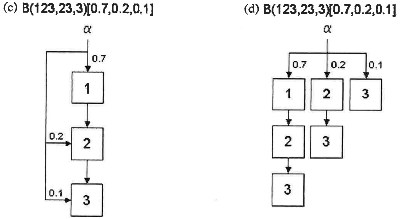
A figure from Burman et al. (2009)

The following diagrams illustrate the graphical representations of multiple testing procedures used by Burman et al.

(a) and (b) A parallel gatekeeping procedure



(c) and (d) A fallback procedure



A figure from Bretz et al. (2009)

And here is an example of a graphical representation of a procedure as defined by Bretz et al.

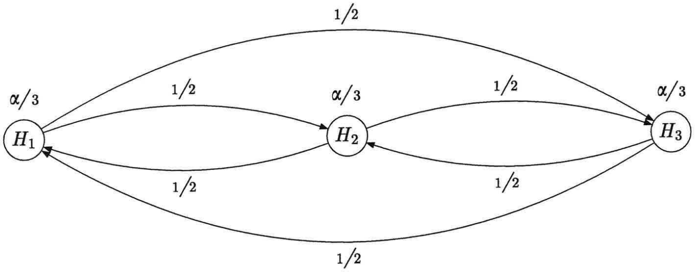


Figure 3. Graphical illustration of the Bonferroni–Holm procedure with $m=3$ hypotheses and initial allocation $\boldsymbol{\alpha}=(\alpha/3, \alpha/3, \alpha/3)$.

Question: How can we apply such a procedure in a group sequential trial?

2.6. Multiple testing within a group sequential design

Maurer & Bretz (*Statist. in Biopharm. Research*, 2013) explain how to carry out tests of multiple hypothesis in a group sequential trial with strong control of FWER.

Consider a multiple testing procedure for hypotheses $H_1, \ldots, H_h$ that involves testing $H_1, \ldots, H_h$ at different significance levels, possibly increasing these levels after other hypotheses are rejected.

Define group sequential tests of each hypothesis with type I error rates equal to the various significance levels that may be applied.

At each analysis, conduct tests of $H_1, \ldots, H_h$ using the boundary points of their group sequential tests for the current analysis.

In doing this, follow the testing hierarchy and "re-cycling rules" to determine the type I error rate of each hypothesis testing boundary.

Stop the study when key conclusions have been reached.

Combining multiple testing and group sequential design

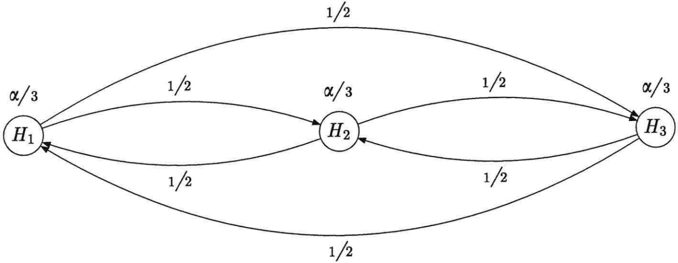


Figure 3. Graphical illustration of the Bonferroni–Holm procedure with $m = 3$ hypotheses and initial allocation $\boldsymbol{\alpha} = (\alpha/3, \alpha/3, \alpha/3)$.

For group sequential implementation of the above multiple testing procedure, we need

GSTs at levels $\alpha/3$, $\alpha/2$ and α

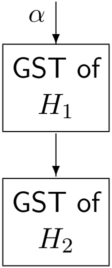
for each of the hypotheses, H_1, H_2 and H_3.

2.7. Testing a secondary endpoint after a sequential test

A correct gatekeeping procedure

We discussed a group sequential trial comparing the effects of two treatments on a primary endpoint. Then, if a positive result is obtained, a secondary endpoint is tested.

In Maurer & Bretz's scheme, we need to specify a level α group sequential test for the secondary endpoint: this test of H_2 will be applied whenever the trial terminates.



The group sequential test of H_1 determines the stopping time for the trial

The group sequential test of H_2 is used for the secondary analysis if and when H_1 is rejected

A correct gatekeeping procedure

Let $Z_{1,1}, \ldots, Z_{1,K}$ be Z-statistics for testing H_1: $\theta_1 \leq 0$ at analyses $1, \ldots, K$.

The group sequential test of H_1 stops at analysis k to

Reject H_1 if $Z_{1,k} \geq b_k$,

Accept H_1 if $Z_{1,k} < a_k$.

Boundary values for the test of H_1 control the type I error rate at level α under $\theta_1 = 0$, i.e.,

$$\sum_{k=1}^{K} P_{\theta_1=0}\{Z_{1,1} \in (a_1, b_1), \ldots, Z_{1,k-1} \in (a_{k-1}, b_{k-1}), Z_{1,k} > b_k\} = \alpha.$$

Suppose this GST stops to reject H_1 at analysis k^* ...

A correct gatekeeping procedure

Let $Z_{2,1}, \ldots, Z_{2,K}$ be Z-statistics for testing H_2: $\theta_2 \leq 0$.

The level α group sequential test of H_2 rejects H_2 at analysis k if $Z_{2,k} \geq c_k$, where

$$\sum_{k=1}^{K} P_{\theta_2=0}\{Z_{2,1} < c_1, \ldots, Z_{2,k-1} < c_{k-1}, Z_{2,k} > c_k\} = \alpha.$$

(The trial's stopping rule is based on the primary endpoint, so we do not need a lower boundary for early acceptance of H_2.)

When the GST of H_1 has rejected H_1 at analysis k^*, we reject H_2 if $Z_{2,k^*} \geq c_{k^*}$.

A gatekeeping procedure *could* reject H_2 if

$$Z_{2,k} \geq c_k \quad \text{for any } k \in \{1, \ldots, K\},$$

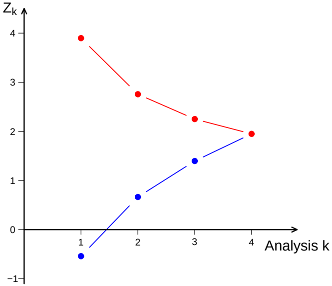
so the FWER is protected conservatively.

Example: Testing primary and secondary endpoints

In a trial comparing two treatments, denote the treatment effects on the primary and secondary endpoints by θ_1 and θ_2.

Suppose the trial is conducted group sequentially, using a Pampallona & Tsiatis test with $\Delta = 0$ for the primary endpoint.

There are 4 analyses, $\alpha = 0.025$ and power is 0.8 at $\theta_1 = 1$.



If H_1: $\theta_1 \leq 0$ is rejected for the primary endpoint at analysis k^*, we test the secondary endpoint: we reject H_2: $\theta_2 \leq 0$ if

$$Z_{2,k^*} \ \geq \ c_{k^*}.$$

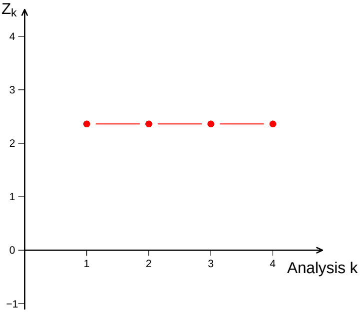
We consider two options for this test of H_2.

Example: Testing primary and secondary endpoints

Consider two options for the group sequential test of H_2.

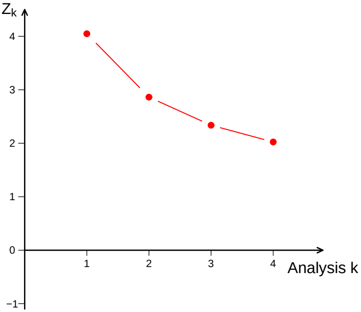
A: Pocock boundary for H_2

$$c_k = 2.361, \quad k = 1, \ldots, 4.$$



B: OBF boundary for H_2

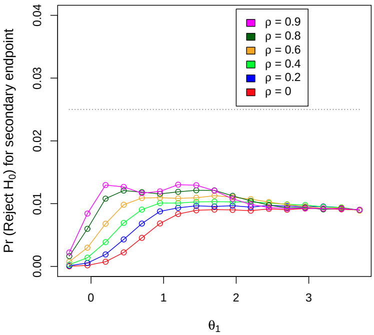
$$c_k = 2.024\sqrt{4/k}, \quad k = 1, \ldots, 4.$$



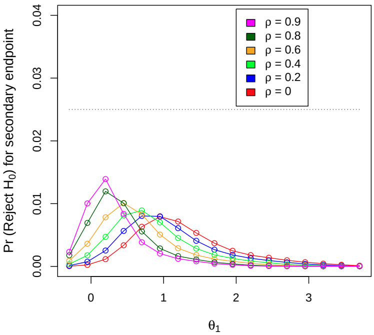
Note: The O'Brien & Fleming boundary requires a very high value of Z_{2,k^*} to reject H_2 if the GST of H_1 stops at the first analysis.

Type I error probability for testing H_2

A: Pocock boundary for H_2



B: OBF boundary for H_2

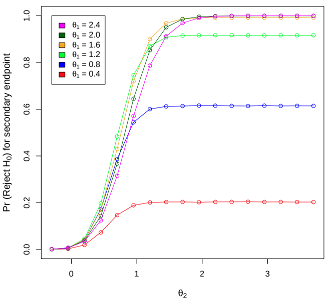


Type I error probabilities are calculated under $\theta_2 = 0$, but they also depend on θ_1 and the correlation, ρ, between the primary and secondary endpoints.

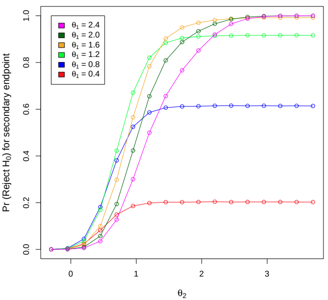
The OBF test of H_2 is particularly conservative when θ_1 is large.

Power for testing H_2, $\rho = 0.25$

A: Pocock boundary for H_2



B: OBF boundary for H_2



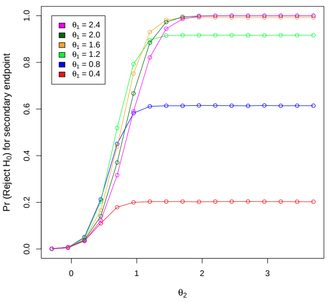
Results are shown for the case that the variance of the secondary response is 0.5 times that for the primary response.

Power is shown as a function of θ_2 for selected values of θ_1.

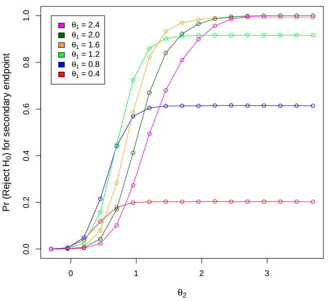
The Pocock boundary for H_2 deals better with the trial's uncertain termination time — which depends significantly on the value of θ_1.

Power for testing H_2, $\rho = 0.5$

A: Pocock boundary for H_2



B: OBF boundary for H_2



Results are shown for the case that the variance of the secondary response is 0.5 times that for the primary response.

Power is shown as a function of θ_2 for selected values of θ_1.

Again, the Pocock boundary for H_2 deals better with the trial's uncertain termination time — which depends significantly on θ_1.

Testing a secondary endpoint: Further options

Conservatism in the overall procedure arises because the test of H_1 may stop at analysis k^* when $Z_{2,k^*} < c_{k^*}$, but

$$Z_{2,k} \geq c_k \text{ for some } k < k^* \text{ or } k > k^*.$$

There are options for reducing conservatism and increasing power:

1. Reject H_2 if $Z_{2,k} \geq c_k$ for some $k < k^*$, even though $Z_{2,k^*} < c_{k^*}$.

   However, ignoring more recent data (and not using the sufficient statistic for θ_2) may detract from the credibility of this decision.

2. Continue the trial to see if $Z_{2,k} \geq c_k$ at a future analysis.

   However, if the primary endpoint is observed for future subjects, the positive result on the primary endpoint could be "lost".

Several authors have considered option (2), retaining a positive outcome for H_1, whatever the additional information about θ_1.

Testing a secondary endpoint: Further options

3. In some cases, the worst case scenario, in which a procedure's maximum FWER occurs, can be identified.

Then, the procedure may be calibrated so that the FWER is equal to the specified level α in this worst case scenario. See:

Glimm, Maurer & Bretz (Stat. in Med., 2010) Hierarchical testing of multiple endpoints in group-sequential trials.

Tamhane, Mehta & Liu (Biometrics, 2010) Testing a primary and a secondary endpoint in a group sequential design.

Tamhane, Wu & Mehta (Stat. in Med., 2012) Adaptive extensions of a two-stage group sequential procedure for testing primary and secondary endpoints (I) unknown correlation between endpoints.

Tamhane, Gou, Jennison, Mehta & Curto (Biometrics, to appear) A gatekeeping test on a primary and a secondary endpoint in a group sequential design with multiple interim looks.

Recapitulation: Multiple hypothesis procedures

- There are many multiple testing schemes to choose from. The most suitable choice will depend on the importance to investigators of rejecting each null hypothesis and the likelihood of each null hypothesis being true or false.
- Graphical representations (SiM papers, 2009) can help investigators to select — and understand — an appropriate multiple testing procedure.
- There are methods available to test multiple hypotheses in a group sequential design AND control the overall type I error probability.
- When testing multiple hypotheses in a group sequential trial design, the key point is to use GSTs as the "testing rules" in the multiple testing scheme: if this is not done correctly, FWER may be inflated.

Part 3. Adaptive clinical trial designs

3.1. Motivation for adaptation in clinical trials

Wall Street Journal, July 2006:

FDA Signals it's Open to Drug Trials that Shift Midcourse

Adaptive designs may allow trials to be adjusted:

- Route more patients to the treatment that seems to work best
- Drop treatments that don't seem to be effective
- Add more of the type of patients ... reacting best to a particular treatment
- Merge two different phases of drug development into one trial

With views from:

Bob O'Neill , FDA

Paul Gallo, Novartis

Tom Fleming, Univ. Washington

Michael Krams, Wyeth

Don Berry, M. D. Anderson Cancer Center

Bruce Turnbull, Cornell University

Adaptation in clinical trials

Adaptation to external factors

Changes in the clinical setting or economic background

Following withdrawal of a competing treatment, a smaller treatment effect is now of clinical interest.

An improved financial position means sponsors can invest more in this trial.

Adaptation to internal factors

Nuisance parameters affecting sample size

In-study estimates of sample variance indicate a greater sample size is needed to achieve the intended power.

Overall failure rates in a survival study are low: higher accrual and longer follow-up are required.

Adaptation in clinical trials

Adaptation to internal factors ...

Safety outcomes

Higher than expected toxicity implies dose should be reduced.

A lower rate of adverse events in the experimental treatment suggests it will suffice to demonstrate non-inferiority, rather than superiority.

Sub-group analyses

The new treatment benefits a particular sub-group: investigators wish to re-define the target population.

Change of endpoint

An alternative endpoint provides better discrimination between treatment groups: investigators wish to re-define the primary endpoint.

Adaptation in clinical trials

Adaptation to internal factors . . .

Response on primary endpoint

Results on the new treatment are good and it is desirable to reach a conclusion as rapidly as possible.

Responses on the new treatment are not as good as anticipated: investigators wish to increase sample size to enhance power at lower effect sizes.

Response-dependent treatment allocation

Interim data suggest one treatment arm could be superior but results are not yet statistically significant. In order to improve treatment of patients in the trial, weight random allocation in favor of the currently superior treatment arm.

Adaptation in clinical trials

A trial with multiple treatments or dose levels

Eliminate weaker treatments as the study progresses.

Using a dose-response model, optimize treatment allocation to learn most efficiently about the best choice of dose level.

Seamless Phase IIb – Phase III trials

Select the best dose level in Phase IIb.

Proceed directly to Phase III and test the treatment at this dose level, eliminating "white space" between phases.

Combine Phase IIb and Phase III data in the final statistical analysis.

FDA Guidance for Industry, February 2010

Adaptive Design Clinical Trials for Drugs and Biologics

Section XI of the document states that adaptations should be pre-specified and access to interim data should be strictly controlled.

> *" . . . there should be comprehensive and prospective, written standard operating procedures (SOPs) that define who will implement the interim analysis and adaptation plan, and all monitoring and related procedures for accomplishing the implementation, providing for the strict control of access to unblinded data (see the DMC guidance)*
>
> *. . . It is likely that the measures defined by the SOPs will be related to the type of adaptation and the potential for impairing study integrity."*

FDA Guidance for Industry, May 2015

Adaptive Designs for Medical Device Clinical Studies

The FDA's guidance for Studies of Medical Devices is similar in spirit to that for Adaptive Trials for Drugs and Biologics.

The phrase "adaptive design" includes group sequential designs with early stopping for efficacy or futility.

The document stresses the importance of:

- Pre-planning and the use of pre-specified adaptation rules,
- Protection of the type I error rate,
- Firewalls to prevent information leakage,
- Assessment of the benefits of adaptive design and weighing of these against more complex logistical requirements,
- The benefits of adaptive designs as a "learning paradigm".

Protecting the type I error probability

The importance of the type I error rate is widely recognised:

- ICH E9 (p. 25)

  "The procedures selected should always ensure that the overall probability of type I error is controlled."

- PhRMA White paper (*J. Biopharmaceutical Statistics*, 2006)

  "The key issue in most contexts is preservation of the type I error rate."

- Pocock & Hughes (*Controlled Clinical Trials*, 1989)

  "Control of type I error is a vital aid to prevent a flood of false positives into the medical literature."

However, for a complex adaptive procedure, it may be difficult to demonstrate control of the type I error rate over the whole of a multi-dimensional null hypothesis.

3.2. Combination tests

Suppose we run a clinical trial adaptively in two stages:

Set the design of Stage 1, then conduct this part of the trial,

Analyse results from Stage 1,

Consider external information, if appropriate.

Set the design of Stage 2, informed by Stage 1 results and external information,

Conduct Stage 2,

Analyse the results from Stage 2.

How can we test a null hypothesis with proper protection of the type I error rate?

Combination tests

Before the trial commences, define the null hypothesis.

Let θ denote the treatment effect *vs* control for a specified form of the treatment, patient population and endpoint.

We test H_0: $\theta \leq 0$ against $\theta > 0$, with type I error rate α at $\theta = 0$.

Define one-sided P-values P_1 and P_2 from hypothesis tests of H_0 based on Stage 1 and Stage 2 data, respectively.

Under $\boldsymbol{\theta = 0}$

$P_1 \sim U(0, 1)$.

Conditionally on all Stage 1 data and the Stage 2 design, $P_2 \sim U(0, 1)$.

Hence, P_1 and P_2 are independent $U(0, 1)$ variates.

The inverse χ^2 combination test

Reference: Bauer & Köhne (*Biometrics*, 1994).

Initial design

Define H_0 and specify the **inverse $\boldsymbol{\chi^2}$ combination test.**

Design Stage 1, fixing the sample size and test statistic.

Stage 1

Observe the one-sided P-value, P_1, based on Stage 1 data.

Design Stage 2 in the light of Stage 1 data.

Stage 2

Observe the P-value, P_2, based on **only** Stage 2 data.

NB: Under $\theta = 0$, $P_1 \sim U(0,1)$, $P_2 \sim U(0,1)$, independent.

Bauer & Köhne's inverse χ^2 combination test

Bauer & Köhne's test rejects H_0 for low values of $P_1\, P_2$.

If $P \sim U(0,1)$, then

$$-\ln(P) \sim \mathsf{Exp}\,(1) = \frac{1}{2}\,\chi^2_2.$$

Thus, under $\theta = 0$,

$$-\ln(P_1\, P_2) \sim \frac{1}{2}\,\chi^2_4.$$

Combining the two P-values in an overall test, we reject H_0 if

$$-\ln(P_1\, P_2) > \frac{1}{2}\,\chi^2_{4,\,1-\alpha}.$$

If $\theta < 0$, then P_1 and P_2 are stochastically larger than $U(0,1)$ random variables and the type I error rate is less than α.

This χ^2 test was originally proposed for combining results of several studies by R. A. Fisher in 1932.

The inverse normal combination test

Initial design

Specify the **inverse normal test** for null hypothesis H_0, with weights w_1 and w_2 where $w_1^2 + w_2^2 = 1$.

Design Stage 1, fixing sample size and test statistic.

Stage 1

Observe the one-sided P-value, P_1, based on Stage 1 data.

Compute $Z_1 = \Phi^{-1}(1 - P_1)$.

Design Stage 2 in the light of Stage 1 data.

Stage 2

Observe the P-value, P_2, based **only** on Stage 2 data.

Compute $Z_2 = \Phi^{-1}(1 - P_2)$.

NB Under $\theta = 0$, $Z_1 \sim N(0, 1)$, $Z_2 \sim N(0, 1)$, independent.

The inverse normal combination test

The combination test rejects H_0 for high values of $w_1 Z_1 + w_2 Z_2$.

Under $\theta = 0$, Z_1 and Z_2 are independent $N(0,1)$, so

$$w_1 Z_1 + w_2 Z_2 \sim N(0,1).$$

Hence, for an overall one-sided test with type I error rate α, we reject H_0 if

$$w_1 Z_1 + w_2 Z_2 > \Phi^{-1}(1-\alpha).$$

If $\theta < 0$, then Z_1 and Z_2 are stochastically smaller than $N(0,1)$ random variables and the type I error rate is less than α.

An attractive property

If w_1 and w_2 are proportional to the square roots of the Stage 1 and Stage 2 sample sizes then $w_1 Z_1 + w_2 Z_2$ is the standard Z-statistic based on the data at the end of Stage 2.

A combination test with more than two groups

The combination test approach can be applied in a group sequential framework (Lehmacher & Wassmer, *Biometrics*, 1999).

For $k = 1, \ldots, K$, define Z_k to be the standardised test statistic based on Stage k data alone, and define cumulative Z-statistics

$$Z(k) \;=\; (w_1 Z_1 + \ldots + w_k Z_k) \,/\, (w_1^2 + \ldots + w_k^2)^{1/2}.$$

Under $\theta = 0$, the sequence $\{Z(k)\}$ follows the canonical joint distribution with

$$\mathrm{Cov}(Z(k_1), Z(k_2)) \;=\; \sqrt{\frac{w_1^2 + \ldots + w_{k_1}^2}{w_1^2 + \ldots + w_{k_2}^2}} \quad \text{for } \; k_1 < k_2.$$

At analysis k, we compare $Z(k)$ with critical values a_k and b_k that define a group sequential test with type I error probability α.

Then, under $\theta = 0$, the probability of rejecting H_0 is exactly α.

3.3. Sample size re-estimation for a response variance

A combination test can be used to protect the type I error rate when a trial's sample size is changed.

Consider a two-treatment comparison in which observations on treatments A and B, respectively, are distributed as

$$X_{Ai} \sim N(\mu_A, \sigma^2) \quad \text{and} \quad X_{Bi} \sim N(\mu_B, \sigma^2).$$

Objective

It is desired to test H_0: $\theta = \mu_A - \mu_B \leq 0$ against $\theta > 0$ with type I error rate α and power $1 - \beta$ at $\theta = \delta$.

In the case of known variance, the sample size formula

$$n = \frac{\{\Phi^{-1}(1-\alpha) + \Phi^{-1}(1-\beta)\}^2 \, 2\,\sigma^2}{\delta^2} \tag{1}$$

gives the required value of n, the sample size per treatment.

However, in practice, only an estimate of σ^2 is usually available.

Sample size re-estimation for a response variance

We can follow an adaptive approach, using an estimate of σ^2 from early trial data to modify the initial choice of sample size.

Initial design

Specify a two-stage adaptive design, using the inverse χ^2 combination rule to test H_0: $\theta \leq 0$ against $\theta > 0$.

Use an initial estimate σ_0^2 in the sample size formula (1) to obtain a sample size of n_0 per treatment.

Stage 1

Conduct Stage 1 with $n_1 = n_0/2$ subjects per treatment.

Observe estimates $\widehat{\theta}_1$, $\hat{\sigma}_1^2$ and the t-statistic t_1 for testing H_0.

Convert t_1 to a one-sided P-value, $P_1 = P_{\theta=0}\{T_{2n_1-2} > t_1\}$.

Sample size re-estimation for a response variance

After Stage 1

Calculate a new Stage 2 sample size of n_2 per treatment arm.

Here, n_2 may be obtained simply by using the new variance estimate $\hat{\sigma}_1^2$ in the original sample size formula.

Or, n_2 might be chosen to give conditional power $1-\beta$ given P_1, assuming $\theta = \widehat{\theta}_1$ and $\sigma^2 = \hat{\sigma}_1^2$.

Stage 2

Calculate the t-statistic t_2 for testing H_0 based on Stage 2 data alone, and obtain the P-value, $P_2 = P_{\theta=0}\{T_{2n_2-2} > t_2\}$.

The inverse χ^2 combination test, which rejects H_0 if

$$-\ln(P_1\,P_2) \;>\; \frac{1}{2}\,\chi^2_{4,\,1-\alpha}$$

has type I error rate exactly equal to α.

Sample size re-estimation for a response variance

The above approach adds to the variety of methods for dealing with an unknown parameter, ϕ, that affects sample size.

Internal pilot:

Wittes & Brittain (*Statistics in Medicine*, 1990) proposed a simple "plug in" of the current estimate $\hat{\phi}$ to update sample size. Bias in the final $\hat{\phi}$ causes a small inflation of the type I error rate.

Unblinded variance estimation:

Friede & Miller (*Applied Statistics*, 2012) show that sample size modification based on an unblinded estimate of σ^2 leads to almost zero type I error rate inflation.

Information monitoring:

Mehta & Tsiatis (*Drug Information Journal*, 2001) "plug in" estimated information in an error spending group sequential design. Typically, this leads to a small inflation of the type I error rate.

Sample size re-estimation for σ^2 in a GST

A Lehmacher and Wassmer K-stage group sequential design:

Weights $w_1, \ldots, w_K$ are specified, where $w_1^2 + \ldots + w_K^2 = 1$.

Let t_k be the t-statistic, on ν_k degrees of freedom, testing H_0: $\theta \leq 0$ vs $\theta > 0$ based on responses in group k alone.

We compute the P-value $P_k = P_{\theta=0}\{T_{\nu_k} > t_k\}$ and define

$$Z_k \; = \; \Phi^{-1}(1 - P_k).$$

At analysis k, we compare the cumulative statistic

$$Z(k) \; = \; (w_1 Z_1 + \ldots + w_k Z_k) \, / \, (w_1^2 + \ldots + w_k^2)^{1/2},$$

with pre-specified critical values a_k and b_k that define a group sequential test with type I error probability α.

Under H_0, $P_k \sim U(0,1)$ and $Z_k \sim N(0,\,1)$ and, hence, this group sequential combination test controls the type I error rate exactly.

An example of sample size modification

Consider a clinical trial studying treatment of heart failure.

Failure rates on control and experimental treatments are p_c and p_t.

From historical data, $p_c \approx 0.25$ and a reduction of 0.05 is desired.

Writing $\theta = p_c - p_t$, it is desired to test H_0: $\theta \leq 0$ against $\theta > 0$ with type I error rate $\alpha = 0.025$ and power $1 - \beta = 0.9$ if $\theta = 0.05$.

Initial design

A Bauer & Köhne two-stage design is specified.

For the above type I error rate and power, assuming $p_c = 0.25$, a fixed sample test needs 1461 subjects per treatment arm.

Stage 1 is planned with 730 subjects per treatment, with a view to re-assessing requirements for the remainder of the study in the light of their responses.

Example: Sample size modification

Stage 1 with 730 subjects per treatment

We observe $\hat{p}_c = 0.253$ and $\hat{p}_t = 0.219$, so $\widehat{\theta} = 0.034$ with standard error 0.0222.

A test of H_0: $\theta \leq 0$ has $Z = 0.034/0.0222 = 1.53$ and P-value

$$P_1 \, = \, 1 - \Phi(1.53) = 0.0629.$$

The overall test will reject H_0 if $-\ln(P_1\, P_2) > 0.5\, \chi^2_{4,\, 0.975} = 5.57.$

Since $-\ln(0.0629) = 2.77$, results thus far are promising — but a positive outcome is by no means certain.

It is learnt that the trial of a competing treatment is unsuccessful.

It is decided to increase the second stage sample size to give a higher probability of a positive outcome under the original alternative, $\theta = 0.05$ — and under smaller effect sizes.

Example: Sample size modification

Planning Stage 2 sample size

p_c	p_t	θ	Stage 2 sample size	Conditional power
0.25	0.22	0.03	750	0.43
			1000	0.51
			1250	0.59
0.25	0.21	0.04	750	0.62
			1000	0.72
			1250	0.80
0.25	0.20	0.05	750	0.78
			1000	0.87
			1250	0.93

Example: Sample size modification

Stage 2

Suppose we decide on 1000 patients per treatment arm in Stage 2.

Stage 2 data (alone) yield $\hat{p}_c = 0.251$ and $\hat{p}_t = 0.221$.

Hence, the Stage 2 data give $\widehat{\theta} = 0.030$ with standard error 0.019.

The test of H_0: $\theta \le 0$ based on Stage 2 data alone has $Z = 0.030/0.019 = 1.58$ and the P-value is

$$P_2 \, = \, 1 - \Phi(1.58) = 0.0570.$$

In the overall test,

$$-\ln(P_1 \, P_2) \, = \, -\ln(0.0629) - \ln(0.0570) \, = \, 2.77 + 2.87 \, = \, 5.64.$$

This is greater than $\frac{1}{2}\,\chi^2_{4,\,0.975} = 5.57$, so H_0 is rejected and it is concluded that the new treatment has a lower failure rate.

Sample size re-estimation in response to $\widehat{\theta}$

The possibility of adaptive design prompted interest in procedures that increase sample size in response to a low interim estimate of the treatment effect.

The objective here is to increase power, recognising that the effect size used in the original power calculation was over-optimistic.

The resulting procedures have an overall maximum possible sample size but, depending on the observed data, the actual sample size can be smaller than this — just like a GST.

JT have compared the properties of adaptive designs and GSTs (*Statistics in Medicine*, 2003 and 2006, *Biometrika*, 2006).

They conclude that familiar group sequential designs can provide almost all the available efficiency gains — whereas some proposals for adaptive designs can be quite inefficient.

Sample size re-estimation in response to $\widehat{\theta}$

Mehta & Pocock (MP, *Statistics in Medicine*, 2011) proposed their "promising zone" trial design, which has the option of adding subjects to increase the (conditional) power for a range of $\widehat{\theta}$ values.

MP describe a trial in which the response is measured 26 weeks after the start of treatment and a large proportion of the total sample will be treated but not yet observed at the interim analysis.

The presence of such "pipeline" subjects imposes a minimum for the total sample size in a design with sample size re-estimation.

Most group sequential tests are designed for the case of an immediate response, with just a few suggestions made for incorporating data from pipeline subjects after a trial is stopped.

However, Hampson & Jennison (*JRSS, B*, 2013) have now proposed group sequential designs for a delayed response.

Sample size re-estimation in response to $\widehat{\theta}$

JT (*Statistics in Medicine*, 2015) discuss Mehta & Pocock's designs and suggest ways to improve their efficiency.

JT define sample size rules that optimize certain sample size and power criteria — achieving the same overall power as MP designs with lower expected sample size.

The designs proposed by JT have smaller increases in sample size but these occur over a wider range than MP's "promising zone".

JT's developments within the MP framework lead, ultimately, to the adaptive version of Hampson and Jennison's Delayed Response GSTs — so we see a convergence of the two approaches.

If fixed group sizes are desired, one can use Hampson and Jennison's non-adaptive delayed response GSTs and their efficiency is still close to the best fully adaptive designs.

3.4. Adaptive trial designs to test multiple hypotheses

There may be changes to key elements during the course of a trial:

Change of treatment definition,

Change of endpoint,

Switching from a test of superiority to a test of non-inferiority.

Or, a trial may proceed in stages with design choices being made at interim analyses

Seamless Phase 2/Phase 3 trial: *Treatment selection, followed by a confirmatory stage*

Enrichment trial: *Possible restriction to patients in a pre-specified sub-group*

Multi-arm multi-stage trial: *Several treatments compared to a control with elimination of poorly performing treatments.*

Adaptive trial designs to test multiple hypotheses

In some of the preceding examples, several null hypotheses are tested at the end of the trial.

In other cases, a single null hypothesis will be tested but the choice of this hypothesis is data dependent — so care is needed to avoid introducing a **selection bias.**

We shall apply multiple testing methods that guarantee protection of the overall type I error rate.

These multiple testing methods will be used in conjunction with combination tests that merge the two sets of data either side of the interim analysis at which an adaptive decision is taken.

Selected references:

Bretz, Schmidli et al. (*Biometrical Journal*, 2006),

Schmidli, Bretz et al. (*Biometrical Journal*, 2006).

Procedures for testing multiple hypotheses

The familywise error rate

Suppose we have h null hypotheses, H_i: $\theta_i \leq 0$ for $i = 1, \ldots, h$.

A procedure's **familywise error rate** when $\theta = (\theta_1, \ldots, \theta_h)$ is

$$P_\theta\{\text{Reject } H_i \text{ for some } i \text{ with } \theta_i \leq 0\}.$$

The familywise error rate is controlled **strongly** at level α if this error rate is at most α for all possible combinations of θ_i values.

Then

$$P_\theta\{\text{Reject any true } H_i\} \;\leq\; \alpha \quad \text{for all } (\theta_1, \ldots, \theta_h).$$

Using such a procedure, the probability of choosing to focus on a parameter θ_{i^*} and then falsely claiming significance for the associated null hypothesis H_{i^*} is at most α.

3.5. Closed testing procedures

Marcus et al. (*Biometrika*, 1976) introduced a **closed testing procedure** which provides strong control of FWER by combining level α tests of each H_i and of intersections of these hypotheses.

Suppose we have null hypotheses H_i, $i = 1, \ldots, h$.

For each subset I of $\{1, \ldots, h\}$, define the intersection hypothesis

$$H_I = \cap_{i \in I} H_i.$$

Construct a level α test of each intersection hypothesis H_I, i.e., a test which rejects H_I with probability at most α whenever all hypotheses specified in H_I are true.

Closed testing procedure

The simple hypothesis H_j: $\theta_j \leq 0$ is rejected overall if, and only if, H_I is rejected for every set I containing index j.

Closed testing procedures

Proof of strong control of familywise error rate

In the closed testing procedure, overall rejection of the simple hypothesis H_j can only occur if H_I is rejected for every set I containing index j.

Let $\tilde{I}$ be the set of indices of all true hypotheses H_i.

Since $H_{\tilde{I}}$ is true, $P\{\text{Reject } H_{\tilde{I}}\} = \alpha$.

For a familywise error to be committed, $H_{\tilde{I}}$ must be rejected.

Hence, the probability of a familywise error is no greater than α.

Testing an intersection hypothesis

Suppose the intersection hypothesis $H_I = \cap_{i \in I} H_i$ is the intersection of m simple hypotheses.

For each $i \in I$, let P_i be the 1-sided P-value for testing H_i.

Denote the ordered values of the P_i by $P_{(1)} \leq P_{(2)} \leq \ldots \leq P_{(m)}$.

There are several ways to test of an intersection hypothesis.

Bonferroni adjustment

The overall P-value for testing H_I is $P_I = m\,P_{(1)}$.

Simes' method (Biometrika, 1986):

The overall P-value for H_I is

$$P_I = \min_{k=1,\ldots,m} (m\,P_{(k)}/k).$$

Bonferroni and Simes' methods

The Bonferroni adjustment is simple, but conservative.

In the definition of Simes' P-value,

$$P_I = \min_{k=1,\ldots,m} (m\, P_{(k)}/k),$$

the term for $k = 1$ is $mP_{(1)}$, i.e., the Bonferroni adjusted P-value.

Other low P-values can reduce the overall result, e.g., if $P_{(2)}$ is only a little higher than $P_{(1)}$ so $P_{(2)}/2 < P_{(1)}$, then this will reduce P_I.

The Simes method is valid — and still slightly conservative — when the P_i are independent or positively dependent.

Such positive dependence arises in a comparison of m treatments with a common control or in tests for a treatment effect in overlapping sub-populations.

Dunnett's method (JASA, 1955)

Suppose m treatments are compared with a control, responses are normal with known variance, and sample sizes on each treatment and the control are equal.

Each null hypothesis H_i says treatment i is no better than control.

We are to test the intersection hypothesis $H_I = \cap_{i \in I} H_i$.

Denote the Z-statistic arising from the test of H_i by Z_i.

When each treatment effect for an $H_i \in H_I$ is zero,

$$Z_i \sim N(0,1), \;\; i \in I, \quad \text{Cov}(Z_i, Z_{i'}) = 0.5, \;\; i \neq i'.$$

The P-value for testing H_I using Dunnett's test is

$$P\{\max_{i \in I} Z_i > z^*\},$$

where z^* is the observed value of $\max_{i \in I} Z_i$, and the probability is under the above multivariate normal distribution for $\{Z_i,\, i \in I\}$.

Using combination tests in a closed testing procedure

Suppose we have null hypotheses H_i, $i = 1, \ldots, h$, and wish to test these in a closed testing procedure with FWER α.

We need to define a level α test for each intersection hypothesis

$$H_I = \cap_{i \in I} H_i$$

— a simple hypothesis H_j is a special case where $I = \{j\}$.

In an adaptive trial with two stages, we can test each H_I by applying a combination test across the stages.

Each stage provides a P-value for H_I and we combine these by a pre-specified method, e.g., "an inverse normal combination test with equal weights".

The P-value for the second stage must be defined before that stage commences, but it may depend on first stage responses.

Adaptive trial designs with multiple hypotheses

The approach of using combination tests within a closed testing procedure is widely applicable.

It provides a way to run trials with the different types of adaptation listed earlier:

Change of treatment definition,

Change of endpoint,

Testing either superiority or non-inferiority,

Seamless Phase 2/Phase 3 trials,

Enrichment trials,

Multi-arm multi-stage trial.

We shall conclude by studying an example of a Phase 3 trial incorporating adaptive treatment selection.

3.6. Example: A survival study with treatment selection

Consider a Phase 3 trial of cancer treatments comparing

Experimental Treatment 1: Intensive dosing

Experimental Treatment 2: Slower dosing

Control treatment

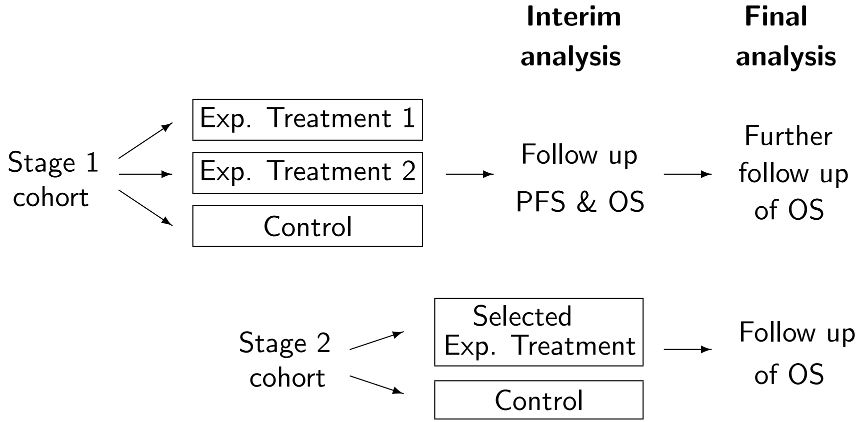
The primary endpoint is Overall Survival (OS).

At an interim analysis, information on OS, Progression Free Survival (PFS), PK measurements and safety will be used to choose between the two experimental treatments.

Note that PFS is useful here as it is more rapidly observed.

After the interim analysis, patients will only be recruited to the selected treatment and the control.

Overall plan of the trial



At the final analysis, we test the null hypothesis that OS on the selected treatment is no better than OS on the control treatment.

Protecting the type I error rate

We shall assume a proportional hazards model for OS with

$$\lambda_1 \; = \; \text{Hazard ratio, Control } \textit{vs} \text{ Exp Treatment 1}$$

$$\lambda_2 \; = \; \text{Hazard ratio, Control } \textit{vs} \text{ Exp Treatment 2}$$

$$\theta_1 \; = \; \log(\lambda_1), \quad \theta_2 \; = \; \log(\lambda_2).$$

We test null hypotheses

$H_{0,1}$: $\theta_1 \leq 0$ vs $\theta_1 > 0$ *(Exp Treatment 1 superior to control)*,

$H_{0,2}$: $\theta_2 \leq 0$ vs $\theta_2 > 0$ *(Exp Treatment 2 superior to control).*

In order to control the "familywise error rate", we require

$$P_{(\theta_1,\theta_2)}\{\text{Reject any true null hypothesis}\} \; \leq \; \alpha.$$

A closed testing procedure

Define level α tests of

$$H_{0,1}\colon\ \theta_1 \le 0,$$

$$H_{0,2}\colon\ \theta_2 \le 0$$

and a level α test of the intersection hypothesis

$$H_{0,12} \ = \ H_{0,1} \cap H_{0,2}\colon\ \ \theta_1 \le 0 \ \text{ and } \ \theta_2 \le 0.$$

Then:

Reject $H_{0,1}$ **overall** *if the above tests reject $H_{0,1}$ and $H_{0,12}$,*

Reject $H_{0,2}$ **overall** *if the above tests reject $H_{0,2}$ and $H_{0,12}$.*

The requirement to reject $H_{0,12}$ compensates for testing multiple hypotheses and the "selection bias" in choosing the treatment to focus on in Stage 2.

Combining data across stages

Consider testing a generic null hypothesis H_0: $\theta \leq 0$ against $\theta > 0$.

Suppose Stage 1 data produce Z_1 where

$$Z_1 \sim N(0,1) \quad \text{if } \theta = 0.$$

After adaptations, Stage 2 data produce Z_2 with *conditional* distribution

$$Z_2 \sim N(0,1) \quad \text{if } \theta = 0.$$

Weighted inverse normal combination test

With pre-specified weights w_1 and w_2 satisfying $w_1^2 + w_2^2 = 1$,

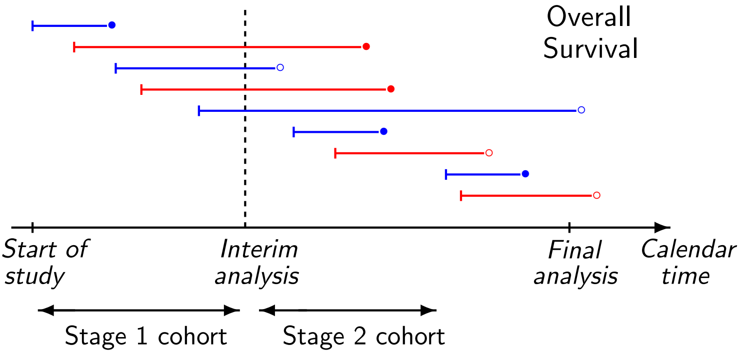
$$Z \;=\; w_1\, Z_1 + w_2\, Z_2 \;\sim\; N(0,1) \quad \text{if } \theta = 0,$$

and Z is stochastically smaller than $N(0,1)$ if $\theta < 0$.

So, for a level α test, we reject H_0 if $Z > \Phi^{-1}(1-\alpha)$.

Properties of log-rank tests

For now, consider Experimental Treatment 1 vs Control.



Key: — Subjects randomised to Exp Treatment 1
— Subjects randomised to Control
• Death observed
○ Censored observation

Properties of log-rank tests

Comparing Experimental Treatment 1 vs Control, define

$$\begin{aligned} S_1 &= \text{Unstandardised log-rank statistic at interim analysis,} \\ \mathcal{I}_1 &= \text{Information for } \theta_1 \text{ at interim analysis} \approx \text{(Number of deaths)}/4 \\ S_2 &= \text{Unstandardised log-rank statistic at final analysis,} \\ \mathcal{I}_2 &= \text{Information for } \theta_1 \text{ at final analysis} \approx \text{(Number of deaths)}/4 \end{aligned}$$

Here, "Number of deaths" refers to the total number of deaths on Experimental Treatment 1 and Control arms only.

Then, approximately,

$$\begin{aligned} S_1 &\sim N(\mathcal{I}_1\,\theta_1,\, \mathcal{I}_1), \\ S_2 - S_1 &\sim N(\{\mathcal{I}_2 - \mathcal{I}_1\}\,\theta_1,\, \{\mathcal{I}_2 - \mathcal{I}_1\}) \end{aligned}$$

and S_1 and $(S_2 - S_1)$ are **independent** (independent increments).

Reference: Tsiatis (*Biometrika*, 1981).

A combination test for survival data

We create Z statistics

Based on data at the interim analysis:

$$Z_1 \; = \; \frac{S_1}{\sqrt{\mathcal{I}_1}},$$

Based on data accrued **between** the interim and final analyses:

$$Z_2 \; = \; \frac{S_2 - S_1}{\sqrt{\mathcal{I}_2 - \mathcal{I}_1}}.$$

If $\theta_1 = 0$, then $Z_1 \sim N(0,1)$ and $Z_2 \sim N(0,1)$ are independent.

If $\theta_1 < 0$, Z_1 and Z_2 are stochastically smaller than this.

So, we can use $Z = w_1\, Z_1 + w_2\, Z_2$ in an inverse normal combination test of $H_{0,1}$: $\theta_1 \leq 0$.

A combination test for survival data

The above distribution theory for logrank statistics of a single comparison requires

$$Z_2 \;=\; \frac{S_2 - S_1}{\sqrt{\mathcal{I}_2 - \mathcal{I}_1}} \;\sim\; N(0,1) \quad \text{under } \theta_1 = 0,$$

regardless of decisions taken at the interim analysis.

Bauer & Posch (*Statistics in Medicine*, 2004) note this implies that the conduct of the second part of the trial should not depend on the prognosis of Stage 1 patients at the interim analysis.

Suppose prognoses are better for patients on Exp Treatment 1 than for those on Control, and the Stage 2 cohort size is reduced while follow up of Stage 1 patients is extended: then, the distribution of Z_2 could be biased upwards.

We shall see there is another potential source of bias in the way the Stage 2 statistic for testing the intersection hypothesis is defined.

Analysing an adaptive survival trial

In applying a Closed Testing Procedure, we require level α tests of

$H_{0,1}$: $\theta_1 \le 0$,

$H_{0,2}$: $\theta_2 \le 0$,

$H_{0,12}$: $\theta_1 \le 0$ and $\theta_2 \le 0$.

Combination tests for these hypotheses are formed from:

	Stage 1 data	*Stage 2 data*
$H_{0,1}$	$Z_{1,1}$	$Z_{2,1}$
$H_{0,2}$	$Z_{1,2}$	$Z_{2,2}$
$H_{0,12}$	$Z_{1,12}$	$Z_{2,12}$

The question is how we should define $Z_{1,1}$, $Z_{2,1}$, etc?

Analysing an adaptive survival trial: Method 1

A natural choice is to:

Base $Z_{1,1}$, $Z_{1,2}$ and $Z_{1,12}$ on data at the interim analysis,

Base $Z_{2,1}$, $Z_{2,2}$ and $Z_{2,12}$ on the additional information accruing between interim and final analyses.

We take $Z_{1,1}$ and $Z_{1,2}$ to be standardised log-rank statistics, and $Z_{2,1}$ and $Z_{2,2}$ their increments between analyses.

For $Z_{1,12}$ we could compare the pooled Exp Tr 1 and Exp Tr 2 patients with the Controls — but, for "consonance" it is better to combine $Z_{1,1}$ and $Z_{1,2}$ through a Simes test or Dunnett test.

If we select Experimental Treatment 1, we no longer test $H_{0,2}$ — we do not need $Z_{2,2}$ and we set $Z_{2,12} = Z_{2,1}$.

Similarly, if we select Experimental Treatment 2, we no longer need $Z_{2,1}$ and we set $Z_{2,12} = Z_{2,2}$.

Method 1: Details

Stage 1 statistics are calculated at the interim analysis:

$Z_{1,1}$ from log-rank test of Exp Tr 1 vs Control

$Z_{1,2}$ from log-rank test of Exp Tr 2 vs Control

$Z_{1,12}$ from pooled log-rank test, or a Simes or Dunnett test.

If Exp Treatment 1 is selected, Stage 2 statistics are

$Z_{2,1}$ from increment in log-rank statistic testing Exp Tr 1 vs Control, combining Stage 1 and Stage 2 cohorts

$Z_{2,12} \quad = Z_{2,1}$.

If Exp Treatment 2 is selected, Stage 2 statistics are

$Z_{2,2}$ from increment in log-rank statistic testing Exp Tr 2 vs Control, combining Stage 1 and Stage 2 cohorts

$Z_{2,12} \quad = Z_{2,2}$.

Method 1: What can go wrong?

The first stage statistics are fine.

Suppose Exp Treatment 1 is selected at the interim analysis.

Now, $Z_{2,1}$ is the increment in the log-rank statistic testing Exp Tr 1 vs Control, combining Stage 1 and Stage 2 cohorts.

$Z_{2,1}$ Issues raised earlier should be considered — might Stage 2 be modified based on interim data in a way that biases $Z_{2,1}$?

Regulators are likely to worry about such possibilities!

$Z_{2,12}$ Setting $Z_{2,12} = Z_{2,1}$ will **definitely** cause bias.

Exp Tr 1 is selected when subjects on this arm have good PFS, so the Exp Tr 1 patients who continue to be followed for OS in Stage 2 are liable to have good prognoses.

This method is almost certain to inflate the type I error rate!!

Method 2: Jenkins, Stone & Jennison (2011)

In constructing a combination test, Method 1 separates data into the parts accrued before and after the interim analysis:

	Z_1	Z_2
Stage 1 cohort	Overall survival (during Stage 1)	Overall survival (during Stage 2)
Stage 2 cohort		Overall survival (during Stage 2)

Instead, we divide the data into the parts from the two cohorts:

Stage 1 cohort	Overall survival (during Stage 1)	Overall survival (during Stage 2)	Z_1
Stage 2 cohort		Overall survival (during Stage 2)	Z_2

Method 2

All patients in the Stage 1 cohort are followed for overall survival up to a fixed time, shortly before the final analysis.

"Stage 1" statistics are based on Stage 1 cohort's **final** OS data

$Z_{1,1}$ from log-rank test of Exp Tr 1 vs Control

$Z_{1,2}$ from log-rank test of Exp Tr 2 vs Control

$Z_{1,12}$ from pooled log-rank test, or a Simes or Dunnett test.

"Stage 2" statistics are based on OS data for the Stage 2 cohort

If Exp Treatment 1 is selected:

$Z_{2,1}$ from log-rank test of Exp Tr 1 vs Control, $Z_{2,12} = Z_{2,1}$

If Exp Treatment 2 is selected:

$Z_{2,2}$ from log-rank test of Exp Tr 2 vs Control, $Z_{2,12} = Z_{2,2}$.

Method 2

Discussion of the design using Method 2

Jenkins, Stone & Jennison (2011) introduced "Method 2" in a design where a choice is made between testing for an effect in the full population or a sub-population.

They stipulated that the amount of follow up for the Stage 1 cohort should be fixed at the outset to avoid any risk of inflating the type I error rate.

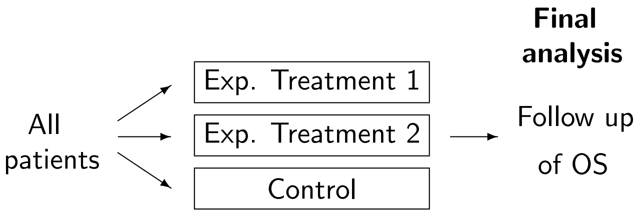
Some adaptive designs allow an early decision based on summaries of "Stage 1" data at an interim analysis.

In our three-treatment design, the statistics $Z_{1,1}$, $Z_{1,2}$ and $Z_{1,12}$ are not known at the time of the interim analysis, so we cannot define a formal stopping rule.

However, with only a little OS data available at the interim analysis, this is not a serious limitation.

Assessing the benefits of an adaptive design

We compare with a non-adaptive trial in which randomisation is to both experimental treatments and control *throughout* the trial.



A closed testing procedure is used to control familywise error rate.

When the total numbers of patients and lengths of follow-up are the same in adaptive and non-adaptive designs,

Does the adaptive design provide higher power?

Are there other advantages?

Assessing the adaptive design: Model assumptions

Overall Survival

	Log hazard ratio
Exp Treatment 1 vs control	θ_1
Exp Treatment 2 vs control	θ_2

Logrank statistics are correlated due to the common control arm.

Progression Free Survival

	Log hazard ratio
Exp Treatment 1 vs control	ψ_1
Exp Treatment 2 vs control	ψ_2

Denote correlation between logrank statistics for OS and PFS by ρ.

Proportional hazards models for both endpoints are not essential (or reasonable?) — the implications for the joint distribution of logrank statistics are what matter.

Assessing the adaptive design: Model assumptions

Log hazard ratios for OS: $\theta_1, \ \theta_2$.

Log hazard ratios for PFS: $\psi_1, \ \psi_2$.

We suppose [... logrank statistics are distributed as if ...]

$\psi_1 = \gamma \times \theta_1$ and $\psi_2 = \gamma \times \theta_2$

Final number of OS events for Stage 1 cohort $= 300$ (over 3 treatment arms)

Number of OS events for Stage 2 cohort $= 300$ (over 2 or 3 treatment arms)

Number of PFS events at interim analysis $= \lambda \times 300$.

When the log hazard ratio is θ, the standardised logrank statistic based on d observed events is, approximately, $N(\theta\sqrt{d/4}, 1)$.

Testing the intersection hypothesis $H_{0,12}$

We have null hypotheses $H_{0,1}$: $\theta_1 \leq 0$ and $H_{0,2}$: $\theta_2 \leq 0$.

In the closed testing procedure we must also test

$$H_{0,12} \;=\; H_{0,1} \cap H_{0,2} : \;\theta_1 \leq 0 \;\; \text{and} \;\; \theta_2 \leq 0.$$

We can test $H_{0,12}$ by pooling the Exp Trt 1 and Exp Trt 2 patients and carrying out a logrank test vs the Control group.

Alternatively we could use a **Simes** test or a **Dunnett** test.

Suppose P_1 and P_2 are the P-values for logrank tests of Exp Trt 1 vs control and Exp Trt 2 vs Control.

The corresponding normal deviates are

$$Z_1 = \Phi^{-1}(1 - P_1) \;\; \text{and} \;\; Z_2 = \Phi^{-1}(1 - P_2).$$

Testing the intersection hypothesis $H_{0,12}$

Simes' test

Given observed values p_1 and p_2 of P_1 and P_2, Simes' test of $H_{0,12}$ yields the P-value

$$\min\left(2\min(p_1, p_2),\ \max(p_1, p_2)\right).$$

Simes' test protects type I error conservatively when P_1 and P_2 are independent or positively associated.

Dunnett's test for comparisons with a common control

If z_1 and z_2 are the observed values of Z_1 and Z_2, the Dunnett test of $H_{0,12}$ yields the P-value

$$P(\max(Z_1, Z_2) \geq \max(z_1, z_2))$$

where (Z_1, Z_2) is bivariate normal with $Z_1 \sim N(0,1)$, $Z_2 \sim N(0,1)$ and $\mathrm{Corr}(Z_1, Z_2) = 0.5$.

Comparing tests of the intersection hypothesis $H_{0,12}$

With $\psi_1 = \theta_1$, $\psi_2 = \theta_2$ and $\rho = 0.6$, we simulated logrank statistics from their large sample distributions under the adaptive design.

We noted

$$
\begin{aligned}
P(1) &= P(\text{Select Treatment 1 and Reject } H_{0,1} \text{ overall}) \\
P(2) &= P(\text{Select Treatment 2 and Reject } H_{0,2} \text{ overall}) \\
E(\text{Gain}) &= \theta_1 \times P(1) \; + \; \theta_2 \times P(2).
\end{aligned}
$$

Here "Gain" is a utility measure which values a positive outcome in proportion to the effect size of the recommended treatment.

In the **adaptive trial design**, we compared use of

Test based on pooled data, Simes' test, and Dunnett's test

for determining Stage 1 $P_{1,12}$ and, hence, $Z_{1,12} = \Phi^{-1}(1 - P_{1,12})$.

Comparing tests of the intersection hypothesis $H_{0,12}$

We compare different forms of $Z_{1,12}$ in an adaptive trial design with

$$\psi_1 = \theta_1, \quad \psi_2 = \theta_2, \quad \lambda = 1, \quad \rho = 0.6, \quad \alpha = 0.025.$$

		$P(1)$			$E(\text{Gain})$		
θ_1	θ_2	Pooled	Simes	Dunnett	Pooled	Simes	Dunnett
0.3	0.0	0.77	0.85	0.86	0.232	0.254	0.259
0.3	0.1	0.78	0.81	0.82	0.238	0.245	0.247
0.3	0.2	0.68	0.68	0.69	0.238	0.237	0.238
0.3	0.25	0.58	0.58	0.58	0.250	0.249	0.249
0.3	0.295	0.48	0.47	0.47	0.275	0.274	0.274

All simulation results are based on 1,000,000 replicates.

The Dunnett test is most effective — it has good power and, unlike the pooled test, is well aligned (consonant) with individual tests of $H_{0,1}$ and $H_{0,2}$.

Comparing adaptive and non-adaptive trial designs

We shall compare designs using a Dunnett test for $H_{0,12}$.

For the non-adaptive design

In the closed testing procedure, we test $H_{0,12}$ at the final analysis.

As for the adaptive design, we found the Dunnett test to give the best power.

If both $H_{0,1}$ and $H_{0,2}$ are rejected, we assume the treatment with higher observed effect size is chosen for registration and marketing.

Accordingly, we define

$$\begin{aligned} P(1) &= P(\widehat{\theta}_1 > \widehat{\theta}_2 \text{ and } H_{0,1} \text{ is rejected overall}) \\ P(2) &= P(\widehat{\theta}_2 > \widehat{\theta}_1 \text{ and } H_{0,2} \text{ is rejected overall}) \\ E(\text{Gain}) &= \theta_1 \times P(1) + \theta_2 \times P(2). \end{aligned}$$

Comparing adaptive and non-adaptive trial designs

We compare designs using a Dunnett test for $H_{0,12}$ with

$$\psi_1 = \theta_1, \quad \psi_2 = \theta_2, \quad \lambda = 1, \quad \rho = 0.6, \quad \alpha = 0.025.$$

		Non-adaptive			Adaptive		
θ_1	θ_2	$P(1)$	$P(2)$	E(Gain)	$P(1)$	$P(2)$	E(Gain)
0.3	0.0	0.78	0.00	0.235	0.86	0.00	0.259
0.3	0.1	0.78	0.01	0.234	0.82	0.02	0.247
0.3	0.2	0.70	0.11	0.234	0.69	0.16	0.238
0.3	0.25	0.60	0.26	0.244	0.58	0.30	0.249
0.3	0.295	0.47	0.43	0.267	0.47	0.44	0.274

Here, $\lambda = 1$ implies there are 300 PFS events at the interim analysis.

The adaptive design has higher $P(1)$ when θ_1 is well above θ_2.

With θ_1 and θ_2 closer, the adaptive design still has higher E(Gain).

Comparing adaptive and non-adaptive trial designs

The adaptive design can only succeed if there is adequate information to select the correct treatment at the interim analysis:

Treatment effects on PFS should be be reliable indicators of treatment effects on OS,

There must be good information on PFS at the interim analysis.

We have investigated varying the parameters γ and λ where

$\psi_1 = \gamma \times \theta_1,\ \psi_2 = \gamma \times \theta_2,$ with $\theta_1 = 0.3$ and $\theta_2 = 0.1$

Final number of OS events for Stage 1 cohort $= 300$ (over 3 arms)

Number of OS events for Stage 2 cohort $= 300$ (over 2 or 3 arms)

Number of PFS events at interim analysis $= \lambda \times 300$.

NB It is quite plausible that γ should be greater than 1.

Comparing adaptive and non-adaptive trial designs

We compare designs with $\theta_1 = 0.3,\ \theta_2 = 0.1,\ \rho = 0.6,\ \alpha = 0.025,$

PFS log hazard ratios: $\psi_1 = \gamma\,\theta_1,\ \psi_2 = \gamma\,\theta_2,$

Number of PFS events at interim analysis $= \lambda \times 300.$

		Non-adaptive			Adaptive		
γ	λ	$P(1)$	$P(2)$	E(Gain)	$P(1)$	$P(2)$	E(Gain)
1.5	1.2				0.88	0.00	0.264
1.2	1.0				0.85	0.01	0.256
1.0	**1.0**	**0.78**	**0.01**	**0.234**	**0.82**	**0.02**	**0.247**
0.9	0.9	for all γ and λ			0.78	0.03	0.238
0.8	0.8	(PFS is not used)			0.74	0.04	0.225
0.7	0.7				0.68	0.05	0.208

Adaptation works well when there is enough PFS information for treatment selection at the interim analysis.

Conclusions about the benefits of an adaptive design

1. The adaptive design offers the chance to select the better treatment and focus on this in the second stage of the trial.

2. Overall, adaptation is beneficial as long as there is sufficient information to make a reliable treatment selection decision.

3. Other evidence may be used in reaching this decision:

   Safety data

   Pharmacokinetic data

   Overall survival

4. In addition to reaching a final decision, both non-adaptive and adaptive trials compare the two forms of treatment: the conclusions from this comparison may be more broadly useful.

Related work

1. Irle & Schäfer (*JASA*, 2012) propose similar adaptive designs for survival data. In place of combination tests, they set critical values for final statistics that preserve the conditional probability of rejecting a null hypothesis after an adaptation.

Arguably, the combination test approach is simpler to explain and easier to implement.

2. Friede et al. (*Statistics in Medicine*, 2011) consider a seamless phase II/III trial design with longitudinal data and treatment selection based on short-term and long-term responses.

Selecting a treatment group based on good short-term results could lead to an upward bias in long-term results for the selected group.

Friede et al. follow a similar approach to Jenkins, Stone & Jennison (2011) and apply a combination test to the long-term response data from the *cohorts* of patients admitted before and after the interim decision point.

Conclusions from the survival example

1. Adaptive designs for trials with survival endpoints can enable interim treatment selection or decisions about the population in which a treatment effect is to be sought.
2. A Closed Testing Procedure can be applied and Combination Tests used to carry out each level α hypothesis test with data from two (or more) stages.
3. The "independent increments" property of the log-rank statistic can fail, and other biases can arise, if design changes at an interim analysis are based on data that are also informative about the later survival of continuing patients.
4. The proposed design avoids this problem. Defining the elements of a combination test in terms of the complete survival data for separate cohorts of patients leads to a valid, and potentially efficient, testing procedure.

Recapitulation: Adaptive clinical trial designs

- It is desirable to adapt a clinical trial design as information becomes available on parameters that were initially unknown.
- Methods are available to create adaptive designs that will protect the overall type I error rate.
- Combination tests allow results from different stages of the trial to be merged.
- Closed Testing Procedures allow tests of multiple hypotheses, or of a single hypothesis selected in a data-dependent manner.
- It should not be assumed that introducing adaptation will automatically make a trial design more efficient.
- Critical appraisal of trial designs is crucial and, where feasible, it is advisable to define an objective function and optimise for this criterion within a chosen class of designs.